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PhysEditWorld: A Large-Scale Dataset Toward Physics-Editable World Models

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Abstract

Recent game world models can synthesize visually plausible, action-conditioned rollouts. However, their interaction behaviors often remain limited to exploratory or wandering trajectories, and physical dynamics are typically learned as implicit correlations from data rather than as controllable variables. This limitation hinders their applicability to authored game environments, where physical rules are deliberately designed and require explicit manipulation. We introduce PhysEditWorld, a multimodal dataset with physical parameters, with a primary focus on gravity in this initial version. At its core, PhysEditWorld is built upon a replay paradigm implemented with a UE5 replay-and-rendering pipeline. Each scenario records a normalized action trace and replays the same initial state, character controller, action sequence, and camera policy under multiple gravity configurations, enabling controlled and attributable physical variation. PhysEditWorld contains 12 cinematic UE5 scenes, over 100 hours of gameplay interactions, and more than 60 million rendered rollout frames. Each sample provides synchronized multimodal signals, including RGB, depth, normals, audio, action traces, camera trajectory, engine states, semantic annotations, and explicit gravity labels. We further conduct initial utility studies on both generative video models and world understanding models, demonstrating that PhysEditWorld enables improved gravity-faithful dynamics modeling, enhances consistency under physical edits, and provides a scalable foundation for controllable world modeling research.
Project page: `https://yizhiqianbi.github.io/physeditworld/`.

1 Introduction

Recent game world models have progressed from visual predictors to interactive generative simulators. Systems such as Genie, DIAMOND, GameNGen, GameGen-X, YUMI, LingBot-World, and Matrix-Game-3.0 show that large generative models can synthesize plausible gameplay trajectories, support

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PhysEditWorld：面向物理可编辑世界模型的大规模数据集

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摘要

近期的游戏世界模型能够合成视觉上可信、以动作作为条件的生成轨迹。然而，其交互行为往往局限于探索或漫游轨迹，物理动力学通常也只是以数据中的隐式相关性被学习，而非作为可控变量。这一局限阻碍了模型用于人工设计的游戏环境，因为其中的物理规则经过刻意设计，需要能够显式操纵。我们提出 PhysEditWorld，一个带物理参数的多模态数据集，其初始版本主要关注重力。PhysEditWorld 的核心是通过 UE5 回放与渲染流水线实现的回放范式。每个场景记录归一化的动作轨迹，并在多种重力配置下，以相同的初始状态、角色控制器、动作序列和相机策略进行回放，从而得到受控且可明确归因的物理变化。PhysEditWorld 包含 12 个电影级 UE5 场景、超过 100 小时的游戏交互，以及超过 6000 万帧渲染轨迹。每个样本提供同步的多模态信号，包括 RGB、深度、法线、音频、动作轨迹、相机轨迹、引擎状态、语义标注和显式重力标签。我们还针对生成式视频模型与世界理解模型开展了初步应用研究，表明 PhysEditWorld 能改善忠实于重力的动力学建模、增强物理编辑下的一致性，并为可控世界建模研究提供可扩展的基础。

项目主页：https://yizhiqianbi.github.io/physeditworld/。

1 引言

近期的游戏世界模型已从视觉预测器发展为交互式生成模拟器。Genie、DIAMOND、GameNGen、GameGen-X、YUMI、LingBot-World 和 Matrix-Game-3.0 等系统表明，大型生成模型能够合成可信的游戏轨迹，并支持

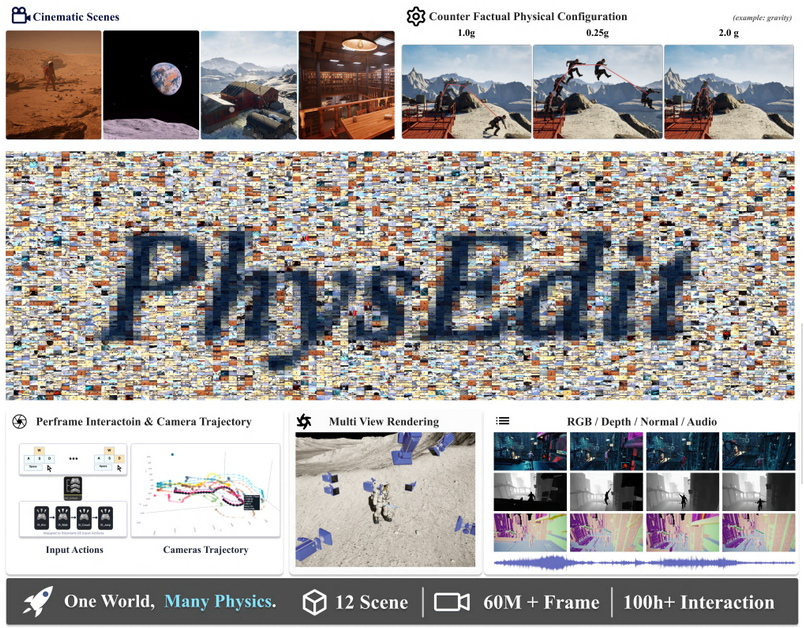

Figure 1: Overview of PhysEditWorld. The dataset contains 12 cinematic UE5 scenes, 100+ hours of gameplay interactions, and 60M+ rendered rollout frames. Recorded interactions are replayed under editable physical configurations and rendered from 8 synchronized camera views, with RGB, depth, normal, audio, action, camera, engine-state, and gravity annotations.

user interaction, or maintain longer-horizon world consistency [1, 2, 3, 4, 5, 6, 7]. However, these models typically learn physics as an implicit regularity of the data distribution. They can imitate how a game usually evolves, but they are not designed to answer a question that is central to game authoring: how should the same scene evolve if a physical rule is edited?

Unlike natural world modeling, where physics is usually treated as a latent constant, game world modeling must eventually support physical laws as editable design variables. Developers routinely tune gravity scale, jump behavior, friction, drag, and wind to shape pacing, difficulty, player feel, and emergent motion. A learned world model that entangles these parameters with appearance and gameplay statistics may generate visually plausible clips, yet still fail as an editable game simulator. In this paper, we focus on gravity as a first measurable step toward editable physics, because it is widely supported across engines and produces observable changes in jump arcs, airtime, fall speed, and object trajectories.

Existing datasets and benchmarks do not directly measure this capability. Game and reinforcement-learning environments such as ALE, Procgen, MineRL, and CARLA provide interactive worlds or human demonstrations, but they are primarily designed for policy learning, navigation, or generalization under fixed rules [8, 9, 10, 11]. World-exploration datasets such as Sekai provide large-scale first-person and drone-view videos with rich annotations and camera trajectories, but are not organized around matched physical interventions [12]. Physics reasoning and video-generation benchmarks such as PHYRE, CLEVRER, Physion, VBench, VideoPhy, PhyGenBench, Physics-IQ, and PhysInOne evaluate intuitive physics, physical plausibility, or physics-aware generation [13, 14, 15, 16, 17, 18, 19, 20]. These resources are valuable, but they do not provide matched gameplay clips in which the scene, interaction trace, and camera policy are held fixed while gravity is explicitly changed.

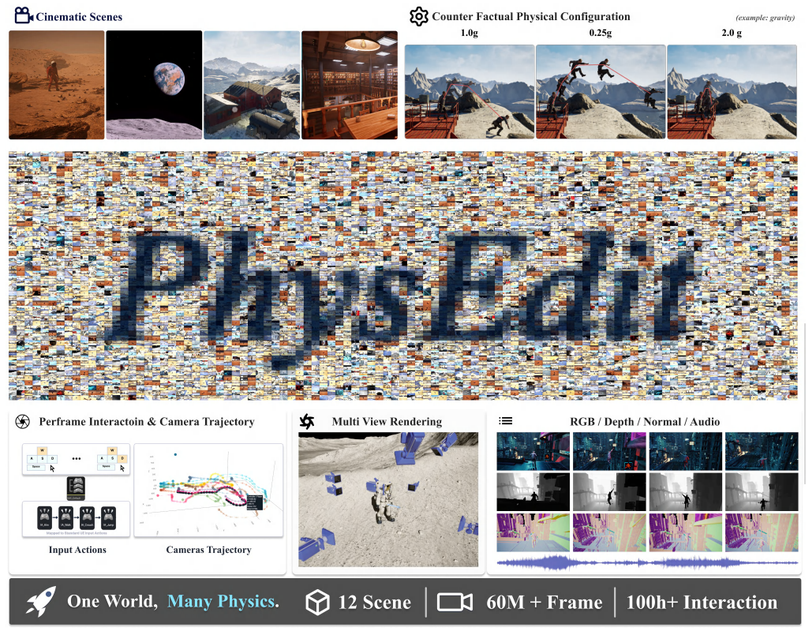

图 1：PhysEditWorld 概览。数据集包含 12 个电影级 UE5 场景、 100+ 小时游戏交互以及 6000 万 + 帧渲染轨迹。记录的交互在可编辑物理配置下回放，并从 8 个同步相机视角渲染，附有 RGB、深度、法线、音频、动作、相机、引擎状态与重力标注。

用户交互，或保持更长时域的世界一致性 [1, 2, 3, 4, 5, 6, 7]。然而，这些模型通常把物理学习为数据分布中的隐式规律。它们可以模仿游戏通常如何演化，却并非为回答游戏创作中的核心问题设计：若编辑一条物理规则，同一场景应如何演化?

自然世界建模通常将物理规律视为潜在常量，而游戏世界建模最终必须把物理定律作为可编辑的设计变量。开发者经常调整重力比例、跳跃行为、摩擦、阻力和风，以塑造节奏、难度、玩家手感及涌现运动。若学习到的世界模型把这些参数与外观及游戏统计规律纠缠在一起，可能生成视觉上可信的片段，却依然不能作为可编辑的游戏模拟器。本文以重力作为实现可编辑物理的第一项可测量目标，因为各类引擎普遍支持重力，而且重力会使跳跃弧线、腾空时长、下落速度及物体轨迹产生可观察的变化。

现有数据集和基准没有直接衡量这种能力。ALE、Procgen、MineRL 和 CARLA 等游戏与强化学习环境提供交互世界或人类示范，但主要面向固定规则下的策略学习、导航或泛化 [8, 9, 10, 11]。Sekai 等世界探索数据集提供大量第一人称和无人机视角视频，带有丰富标注与相机轨迹，但并未围绕配对物理干预组织数据 [12]。PHYRE、CLEVRER、Physion、VBench、VideoPhy、PhyGenBench、Physics-IQ 和 PhysInOne 等物理推理与视频生成基准，评估直觉物理、物理合理性或物理感知生成[13, 14, 15, 16, 17, 18, 19, 20]。这些资源很有价值，但不提供这样的配对游戏片段：固定场景、交互轨迹和相机策略，同时显式改变重力。

We introduce **PhysEditWorld**, a multimodal dataset for physics-editable game world modeling. PhysEditWorld contains 12 cinematic UE5 scenes and over 60M rendered frames, with matched rollouts designed for gravity-conditioned generation and evaluation. The dataset is built around *comparability*: each replay group fixes the authored scene, initial state, action trace, character controller, and camera policy, then reruns the scenario under different gravity configurations. The pipeline records synchronized first-person RGB, third-person RGB, depth, normal maps, action traces, semantic captions, engine states, and gravity annotations. Because non-gravity factors are controlled within a replay group, the resulting motion differences can be evaluated as responses to the physical intervention.

PhysEditWorld enables evaluation beyond visual plausibility by testing whether generated rollouts are faithful to edited gravity. We study this capability in gravity-conditioned generation, first-person world-model rollouts, and video-language gravity inference. Across representative backbones, we find that current models can maintain visual realism, but often under-express gravity-sensitive motion or confuse the relative ordering of gravity levels. This suggests that editable physics remains a missing capability in current game world models.

Our contributions are as follows.

- We introduce **PhysEditWorld**, a large-scale multimodal dataset for physics-editable game world modeling. To the best of our knowledge, it is the first dataset organized around matched gameplay rollouts with explicit gravity interventions.
- We develop a UE5 **replay-and-rendering pipeline** that automatically replays the same scene, initial state, character controller, action trace, and camera policy under controlled physics configurations.
- We conduct dataset utility studies showing that PhysEditWorld can be used to evaluate and improve gravity awareness in generative video models, game world models, and video-language models.

2 Related work

Game world models. Game world models have evolved from single-game neural simulators to open-domain interactive video generators. Early works such as GameGAN, Playable Video Generation, Playable Environments, and Promptable Game Models studied controllable neural simulation and interactive video manipulation from gameplay or video data [21, 22, 23, 24, 25]. Recent systems, including Genie, Oasis, DIAMOND, GameNGen, GameGen-X, GameFactory, MineWorld, YUME, LingBot-World, Matrix-Game 3.0, and related interactive video world models, scale action-conditioned generation, open-world exploration, long-horizon consistency, ID consistency, and real-time interaction [1, 26, 2, 3, 4, 27, 28, 7, 5, 6, 29, 30]. While these models learn rich visual and interaction dynamics, physical rules are usually absorbed as implicit dataset regularities rather than exposed as editable variables. PhysEditWorld addresses this gap with explicit gravity annotations and matched replays under fixed scene, action, controller, and camera conditions.

Video and world-model datasets. Existing game and world-model datasets mainly emphasize scale, action control, exploration coverage, or state supervision. MineRL, VPT, and MineDojo provide Minecraft demonstrations, video pretraining data, simulator environments, or task suites for embodied agent learning [10, 31, 32]. OGameData/GameGen-X and GF-Minecraft/GameFactory target open-world or action-controllable game video generation [4, 27]; Sekai and YUME focus on world exploration and interactive generation [12, 7]; and WildWorld and MultiWorld add explicit state, multi-agent, or multi-view supervision [33, 34]. Related resources also study high-DoF action-to-video control or physics-related gameplay failures [35, 36]. However, these datasets are not organized around matched physical interventions. PhysEditWorld instead replays the same authored scene, interaction trace, character controller, and camera policy under different gravity configurations, making the physical edit directly comparable.

Physics benchmarks and controllable generation. Many benchmarks study physical understanding from visual data. ShapeStacks and ADEPT probe object stability and expectation violation, while PHYRE, I-PHYRE, IntPhys, IntPhys2, CLEVRER, CATER, and Physion evaluate intuitive physics, compositional actions, causal reasoning, intervention, and future prediction [37, 38, 13, 39, 40, 41, 14, 42, 15]. CoPhy, ComPhy, CRIPP-VQA, ContPhy, CausalVQA,

我们提出 PhysEditWorld，一个用于物理可编辑游戏世界建模的多模态数据集。它包含 12 个电影级 UE5 场景与超过 6000 万帧渲染图像，其配对轨迹专为重力条件生成和评估设计。数据集围绕可比较性构建：每个回放组固定人工设计的场景、初始状态、动作轨迹、角色控制器和相机策略，然后在不同重力配置下重新运行场景。流水线记录同步的第一人称 RGB、第三人称 RGB、深度、法线图、动作轨迹、语义描述、引擎状态和重力标注。由于回放组内的非重力因素受到控制，产生的运动差异可以作为对物理干预的响应进行评估。

PhysEditWorld 通过检验生成轨迹是否忠实于编辑后的重力，实现超越视觉合理性的评估。我们在重力条件生成、第一人称世界模型生成轨迹以及视频语言重力推断中研究这种能力。在代表性骨干模型上发现，当前模型能够保持视觉逼真度，却经常不足以表现重力敏感运动，或混淆重力大小的相对顺序。这表明，可编辑物理仍是当前游戏世界模型欠缺的能力。

我们的贡献如下。

- 我们提出 PhysEditWorld，一个用于物理可编辑游戏世界建模的大规模多模态数据集。据我们所知，它是首个围绕带显式重力干预的配对游戏生成轨迹组织的数据集。
- 我们开发了 UE5 回放与渲染流水线，在受控物理配置下，自动回放相同场景、初始状态、角色控制器、动作轨迹和相机策略。
- 我们开展数据集应用研究，表明 PhysEditWorld 可用于评估并改善生成视频模型、游戏世界模型和视频语言模型的重力感知能力。

2 相关工作

游戏世界模型。游戏世界模型已从单一游戏的神经模拟器发展为开放领域的交互式视频生成器。GameGAN、Playable Video Generation、Playable Environments 和 Promptable Game Models 等早期工作利用游戏或视频数据，研究可控神经模拟与交互式视频操纵 [21, 22, 23, 24, 25]。Genie、Oasis、DIAMOND、GameNGen、GameGen-X、GameFactory、MineWorld、YUME、LingBot-World、Matrix-Game 3.0 及相关交互视频世界模型等近期系统，扩展了动作条件生成、开放世界探索、长时域一致性、身份一致性和实时交互的规模 [1, 26, 2, 3, 4, 27, 28, 7, 5, 6, 29, 30]。尽管这些模型学到了丰富的视觉与交互动力学，物理规则通常被吸收为数据集的隐式规律，而没有作为可编辑变量显式暴露。PhysEditWorld 通过显式重力标注，以及在固定场景、动作、控制器和相机条件下的配对回放，填补这一空白。

视频与世界模型数据集。现有游戏和世界模型数据集主要强调规模、动作控制、探索覆盖度或状态监督。MineRL、VPT 和 MineDojo 为具身智能体学习提供 Minecraft 示范、视频预训练数据、模拟器环境或任务套件 [10, 31, 32]。OGameData/GameGen-X 和 GF-Minecraft/GameFactory 面向开放世界或动作可控的游戏视频生成 [4, 27]；Sekai 和 YUME 关注世界探索与交互生成 [12, 7]；WildWorld 和 MultiWorld 增加显式状态、多智能体或多视角监督 [33, 34]。相关资源还研究高自由度的动作到视频控制，以及游戏中的物理失效 [35, 36]。但这些数据集并未围绕配对物理干预组织。PhysEditWorld 则在不同重力配置下，回放同一人工设计的场景、交互轨迹、角色控制器与相机策略，使物理编辑可以直接比较。

物理基准与可控生成。许多基准研究从视觉数据理解物理。ShapeStacks 和 ADEPT 探究物体稳定性与预期违背；PHYRE、I-PHYRE、IntPhys、IntPhys2、CLEVRER、CATER 和 Physion 评估直觉物理、组合动作、因果推理、干预与未来预测 [37, 38, 13, 39, 40, 41, 14, 42, 15]。CoPhy、ComPhy、CRIPP-VQA、ContPhy、CausalVQA,

and QuantiPhy further study counterfactual dynamics, hidden physical properties, causal alternatives, and quantitative physical quantities [43, 44, 45, 46, 47, 48]. Video-generation benchmarks such as VBench, VBench-2.0, VideoPhy, PhyGenBench, Physics-IQ, PhysInOne, WorldScore, and NewtonRewards evaluate physical plausibility, intrinsic faithfulness, world-generation quality, or Newtonian motion consistency [16, 49, 17, 18, 19, 20, 50, 51, 52, 53]. Recent controllable generation methods condition on force, torque, force fields, physical parameters, Newtonian dynamics, or learned physical priors [54, 55, 56, 57, 58, 59, 60]. These works provide important reasoning, evaluation, and generation tools, but they do not provide interactive game rollouts where one physical rule is edited while scene content and interaction are held fixed.

Synthetic simulation platforms. Simulation platforms enable controllable data generation and ground-truth annotation. ALE, Procgen, MineRL, and CARLA support standardized environments for games, reinforcement learning, and driving [8, 9, 10, 11]. AI2-THOR, RoboTHOR, ProcTHOR, Habitat, Gibson, iGibson, TDW, Kubric, VirtualHome, and BEHAVIOR-1K provide controllable 3D simulation with rich sensor outputs, physical interaction, programmatic actions, or scalable embodied-AI environments [61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71]. Robotics simulators such as SAPIEN, RLBench, ManiSkill2, and Isaac Gym support articulated-object interaction, manipulation, and high-throughput physics simulation [72, 73, 74, 75]. UnrealCV connects Unreal Engine to computer-vision pipelines, while UnrealZoo scales UE environments for embodied AI [76, 77]. PhysEditWorld builds on UE but targets a different protocol: matched replay under edited gravity.

Table 1: Comparison with related datasets and data-generation resources. RGB-D-N-A denotes synchronized RGB, depth, normal, and audio. Edit phys. denotes whether physical attributes are explicitly editable; PhysEditWorld focuses on gravity in the current release. Camera ann. denotes camera parameters or trajectories. Action/control denotes recorded actions, controls, or interaction traces. ✓: yes, ✗: no, △: partial.

Resource	Multi-view	RGB-D-N-A	Edit phys.	Camera ann.	Action
ALE [8]	✗	✗	✗	✗	✓
Procgen [9]	✗	✗	✗	✗	✓
MineRL [10]	✗	✗	✗	△	✓
Sekai [12]	△	△	✗	✓	✗
GameFactory / GF-Minecraft [27]	✗	✗	✗	✗	✓
OGameData / GameGen-X [4]	△	✗	✗	✓	△
CLEVRER [14]	✗	✗	✗	✗	✗
Physion [15]	✗	△	✗	△	✗
VBench [16]	✗	✗	✗	✗	✗
PhyGenBench [18]	✗	✗	✗	✗	✗
VideoPhy [17]	✗	✗	✗	✗	✗
Physics-IQ [19]	✗	✗	✗	✗	✗
PhysInOne [20]	✓	△	✗	△	✗
PhysEditWorld	✓	✓	✓	✓	✓

3 Physics Edit Dataset

PhysEditWorld is a large-scale multimodal dataset for gravity-editable game world modeling. It contains 12 cinematic-quality UE5 scenes, more than 100 hours of human gameplay interactions, and over 60 million rendered rollout frames across multiple gravity configurations and synchronized camera views. All videos are rendered at 30 FPS and 1280×720 resolution. Unlike many physics-video datasets that focus on short, isolated physical events, PhysEditWorld is built from interactive game scenarios, enabling gravity effects to be studied in action-conditioned world-model rollouts.

Each rollout provides synchronized RGB video, depth maps, surface normals, audio when available, semantic captions, action traces, camera parameters, engine-state logs, and explicit gravity labels. The engine logs include camera trajectories, character states, object states, and relevant physical variables exported in UE5's native world coordinate system and physical scale. The basic unit is a *matched replay group*: the scene, initial state, character controller, action trace, and camera policy are fixed, while only the gravity configuration changes. This structure makes gravity-dependent effects

和 QuantiPhy 进一步研究反事实动力学、隐含物理属性、因果替代情况及定量物理量 [43, 44, 45, 46, 47, 48]。VBench、VBench-2.0、VideoPhy、PhyGenBench、Physics-IQ、PhysInOne、WorldScore 和 NewtonRewards 等视频生成基准，评估物理合理性、内在忠实性、世界生成质量或牛顿运动一致性 [16, 49, 17, 18, 19, 20, 50, 51, 52, 53]。近期的可控生成方法以力、力矩、力场、物理参数、牛顿动力学或学习到的物理先验为条件 [54, 55, 56, 57, 58, 59, 60]。这些工作提供了重要的推理、评估和生成工具，但没有提供这样的交互式游戏生成轨迹：只编辑一条物理规则，同时保持场景内容与交互不变。

合成模拟平台。模拟平台能够实现可控数据生成和真实值标注。ALE、Procgen、MineRL 和 CARLA 为游戏、强化学习与自动驾驶提供标准化环境 [8, 9, 10, 11]。AI2-THOR、RoboTHOR、ProcTHOR、Habitat、Gibson、iGibson、TDW、Kubric、VirtualHome 和 BEHAVIOR-1K 提供可控三维模拟，具备丰富的传感器输出、物理交互、程序化动作或可扩展的具身人工智能环境 [61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71]。SAPIEN、RLBench、ManiSkill2 和 Isaac Gym 等机器人模拟器支持关节物体交互、操作以及高吞吐物理模拟 [72, 73, 74, 75]。UnrealCV 将 Unreal Engine 接入计算机视觉流水线，UnrealZoo 则扩展了用于具身人工智能的 UE 环境规模 [76, 77]。PhysEditWorld 同样基于 UE，但采用不同的协议：在编辑后的重力条件下进行配对回放。

表 1：与相关数据集及数据生成资源的比较。RGB-D-N-A 表示同步的 RGB、深度、法线和音频。物理编辑表示物理属性是否显式可编辑；当前 PhysEditWorld 版本聚焦重力。相机标注表示相机参数或轨迹。动作／控制表示记录的动作、控制或交互轨迹。√：有，：无， △：部分。

资源	多视角	RGB-D-N-A	物理编辑	相机标注	动作
ALE [8]	✗	✗	✗	✗	✓
Procgen [9]	✗	✗	✗	✗	✓
MineRL [10]	✗	✗	✗	△	✓
Sekai [12]	△	△	✗	✓	✗
GameFactory / GF-Minecraft [27]	✗	✗	✗	✗	✓
OGameData / GameGen-X [4]	△	✗	✗	✓	△
CLEVRER [14]	✗	✗	✗	✗	✗
Physion [15]	✗	△	✗	△	✗
VBench [16]	✗	✗	✗	✗	✗
PhyGenBench [18]	✗	✗	✗	✗	✗
VideoPhy [17]	✗	✗	✗	✗	✗
Physics-IQ [19]	✗	✗	✗	✗	✗
PhysInOne [20]	✓	△	✗	△	✗
PhysEditWorld	✓	✓	✓	✓	✓

3 物理编辑数据集

PhysEditWorld 是用于重力可编辑游戏世界建模的大规模多模态数据集。它包含 12 个电影级 UE5 场景、超过 100 小时的人类游戏交互，以及跨多种重力配置和同步相机视角的超过 6000 万帧渲染轨迹。所有视频均以 30 FPS、 1280×720 分辨率渲染。不同于聚焦短暂、孤立物理事件的许多物理视频数据集，PhysEditWorld 基于交互式游戏场景构建，使研究者能在动作条件世界模型生成轨迹中研究重力效应。

每条生成轨迹提供同步的 RGB 视频、深度图、表面法线、可用时的音频、语义描述、动作轨迹、相机参数、引擎状态日志和显式重力标签。引擎日志包括相机轨迹、角色状态、物体状态，以及以 UE5 原生世界坐标系与物理尺度导出的相关物理变量。基本单位是配对回放组：固定场景、初始状态、角色控制器、动作轨迹和相机策略，只改变重力配置。这一结构使重力相关效应

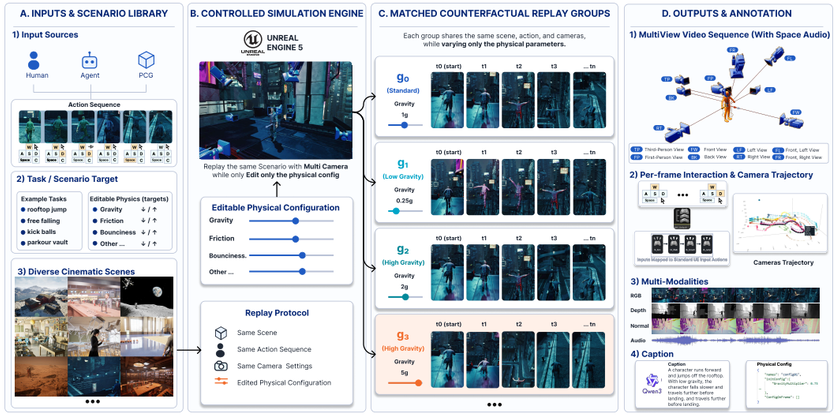

Figure 2: Overview of the PhysEditWorld data pipeline. A) Input sources and scenario library provide human-controlled, agent-generated, or scripted action sequences in diverse cinematic UE5 scenes. B) A controlled UE5 simulation engine replays the same scenario while editing only the physical configuration. C) Matched replay groups share the same scene, action sequence, character controller, and camera policy while varying gravity. D) The pipeline exports synchronized multi-view videos, spatial audio, per-frame interaction and camera trajectories, multimodal render passes, captions, and physical annotations.

such as jump height, airtime, fall speed, landing timing, camera displacement, and object motion directly comparable across variants.

Table 1 summarizes the advantage of PhysEditWorld over prior game, world-model, and physics-oriented datasets. Physics-video and physical-reasoning benchmarks such as CLEVRER, Physion, VBench, VideoPhy, PhyGenBench, Physics-IQ, and PhysInOne provide useful tests of physical plausibility, intuitive physics, or physics-aware generation, but generally lack interactive action traces and matched physical interventions [14, 15, 16, 17, 18, 19, 20]. Game and world-modeling datasets such as ALE, Procgen, MineRL, Sekai, GameFactory/GF-Minecraft, and OGameData/GameGen-X provide action control, visual scale, or camera information to varying degrees, but typically operate under fixed physical rules [8, 9, 10, 12, 27, 4]. In contrast, PhysEditWorld combines multi-view rendering, RGB-depth-normal-audio supervision, explicit editable gravity, camera annotations, and action/control traces, making it possible to evaluate whether a model changes world dynamics consistently under the same scene, action sequence, and camera policy.

4 Physics Data Pipeline

PhysEditWorld is generated by a UE5 replay-and-rendering pipeline integrated into the standard game-development workflow. Rather than reconstructing scenes in a separate simulator, the pipeline operates directly on artist-ready UE5 levels through an in-editor plug-in, converting authored game content into data-production-ready scenarios with registered scenes, validated assets, controller bindings, camera policies, and replay settings. It then builds normalized action sequences from human-controlled, agent-generated, or scripted inputs and replays them through the same UE5 gameplay stack under controlled physical configurations. The final stage performs synchronized rendering, runtime logging, data organization, and post-hoc VLM annotation, producing aligned rollouts with semantic descriptions.

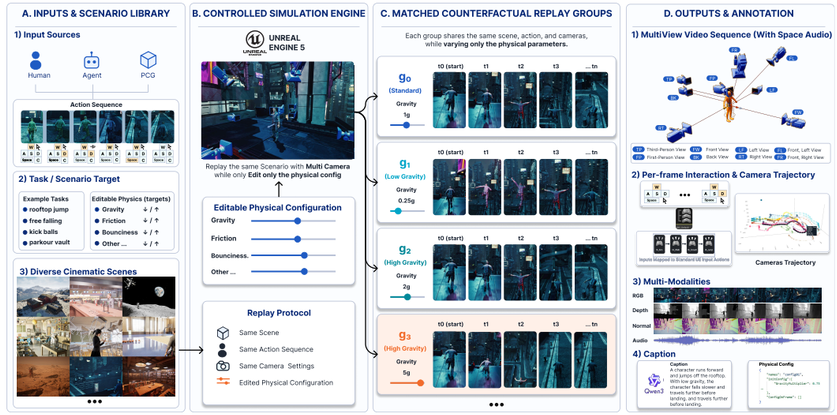

图 2：PhysEditWorld 数据流水线概览。A）输入来源与场景库在多样的电影级 UE5 场景中提供人工控制、智能体生成或脚本化动作序列。B）受控 UE5 模拟引擎回放同一场景，仅编辑物理配置。C）配对回放组共享相同场景、动作序列、角色控制器与相机策略，仅改变重力。D）流水线导出同步的多视角视频、空间音频、逐帧交互与相机轨迹、多模态渲染通道、描述文本和物理标注。

如跳跃高度、腾空时长、下落速度、着陆时序、相机位移和物体运动，都可在不同变体间直接比较。

表 1 概述了 PhysEditWorld 相比既有游戏、世界模型和物理数据集的优势。CLEVRER、Physion、VBench、VideoPhy、PhyGenBench、Physics-IQ 和 PhysInOne 等物理视频与物理推理基准，可有效测试物理合理性、直觉物理或物理感知生成，但通常缺少交互式动作轨迹和配对物理干预 [14, 15, 16, 17, 18, 19, 20]。ALE、Procgen、MineRL、Sekai、GameFactory/GF-Minecraft 和 OGameData/GameGen-X 等游戏与世界建模数据集，以不同程度提供动作控制、大规模视觉数据或相机信息，但通常运行于固定物理规则之下 [8, 9, 10, 12, 27, 4]。相比之下，PhysEditWorld 结合多视角渲染、RGB—深度—法线—音频监督、显式可编辑重力、相机标注以及动作／控制轨迹，因此可以评估：在相同场景、动作序列与相机策略下，模型是否能一致地改变世界动力学。

4 物理数据流水线

PhysEditWorld 由集成到标准游戏开发流程中的 UE5 回放与渲染流水线生成。该流水线不在独立模拟器中重建场景，而是通过编辑器插件直接处理供艺术人员使用的 UE5 关卡，将人工制作的游戏内容转换为可生产数据的场景，包含已注册场景、经验证资产、控制器绑定、相机策略和回放设置。随后，它从人工控制、智能体生成或脚本输入中构建归一化动作序列，并在受控物理配置下通过同一 UE5 游戏系统进行回放。最后执行同步渲染、运行时日志记录、数据组织以及事后视觉语言模型标注，生成附带语义描述的对齐轨迹。

4.1 Scenario and input construction

The pipeline starts from artist-ready UE5 levels and converts them into replayable scenarios through the in-editor plug-in. Each scenario registers the authored level, interactable assets, character setup, collision configuration, camera anchors, and replay settings required for controlled simulation. Scenario preparation emphasizes physical visibility and replay stability: artists select events such as jumps, falls, object interactions, and height-varying traversal, and validate that the corresponding assets, controllers, spawn points, and camera anchors support reproducible replay.

For each prepared scenario, the pipeline constructs action sequences from human-controlled, agent-generated, or scripted inputs. Instead of recording raw device events, it records semantic Input Action sequences from UE5's Enhanced Input System, including movement axes, jump commands, camera deltas, and key or button states. This representation preserves the intent of the interaction while avoiding hardware-specific confounders such as keyboard layout, mouse sensitivity, controller mapping, or platform-dependent event APIs.

4.2 Controlled UE5 simulation

To scale data production within standard game-development workflows, PhysEditWorld uses an in-editor DataFactory plug-in rather than a separate simulator. The plug-in operates on native UE5 abstractions, including authored levels, character controllers, Enhanced Input actions, camera components, collision volumes, and gameplay logic. It provides unified tools for scene registration, replay specification, controller binding, camera setup, physical-parameter editing, validation, and batch execution.

During simulation, the normalized action sequence is injected back into the same UE5 gameplay stack used during capture. Physical configurations are applied as explicit simulation parameters, while the authored scene, controller logic, input sequence, and camera policy remain fixed. This design separates interaction capture from physical intervention and allows controlled physical edits without recollecting behavior or modifying the authored level.

4.3 Matched replay expansion

Let A denote the prepared UE5 scenario, S the normalized action sequence, M the character controller, θ the editable physical configuration, and π_c the camera policy. A rollout is generated as

$$x = F(A, S, M, \theta, \pi_c). \tag{1}$$

Replay expansion keeps (A, S, M, π_c) fixed and reruns the same scenario under different values of θ. This construction makes the physical edit the controlled variable, so that differences across variants can be attributed to the edited simulation parameter rather than to changes in scene content, input behavior, character setup, or camera specification.

The action sequence is treated as scene-bound because its semantics depend on local geometry, obstacles, and interaction targets. The pipeline therefore expands validated scenario-action pairs over compatible physical configurations and character setups, producing matched variants while preserving the original authored environment and interaction intent.

4.4 Synchronized export and annotation

Each replay is exported through Movie Render Queue and runtime logging. Movie Render Queue produces rendered observation streams and auxiliary render passes, while the runtime logger records time-aligned interaction, camera, state, and physical-configuration metadata. All exported records are indexed by a shared rollout identity and frame timeline, enabling deterministic alignment between rendered observations and engine-side logs.

After rendering, the pipeline performs data organization, post-hoc semantic annotation, and quality filtering. We annotate each rollout with a per-clip caption generated by Qwen3-VL-8B-Instruct [78]; the captioning model observes sampled rendered frames only and does not receive simulator metadata. Replays with rendering failures, broken synchronization, severe camera clipping, unstable simulation, or divergence not attributable to the intended physical edit are discarded.

4.1 场景与输入构建

流水线从可供艺术人员使用的 UE5 关卡开始，通过编辑器插件将其转换为可回放场景。每个场景注册人工设计的关卡、可交互资产、角色设置、碰撞配置、相机锚点，以及受控模拟所需的回放设置。场景准备强调物理现象可见性与回放稳定性：艺术人员选择跳跃、下落、物体交互和高度变化穿行等事件，并验证相关资产、控制器、出生点与相机锚点能支持可复现回放。

对于每个准备好的场景，流水线从人工控制、智能体生成或脚本输入构建动作序列。它不记录原始设备事件，而是记录 UE5 Enhanced Input System 的语义 Input Action 序列，包括移动轴、跳跃命令、相机增量和键盘或按钮状态。这种表示保留交互意图，同时避免键盘布局、鼠标灵敏度、手柄映射或平台事件 API 等硬件相关混杂因素。

4.2 受控 UE5 模拟

为在标准游戏开发流程中扩大数据生产规模，PhysEditWorld 使用编辑器内的 DataFactory 插件，而非独立模拟器。插件操作原生 UE5 抽象，包括人工设计的关卡、角色控制器、EnhancedInput 动作、相机组件、碰撞体积和游戏逻辑。它为场景注册、回放规格、控制器绑定、相机设置、物理参数编辑、验证及批量执行提供统一工具。

模拟时，归一化动作序列被重新注入采集时使用的同一 UE5 游戏系统。物理配置作为显式模拟参数应用，而人工设计的场景、控制器逻辑、输入序列与相机策略保持固定。这一设计将交互采集与物理干预分离，无需重新收集行为或修改人工设计的关卡，即可实施受控物理编辑。

4.3 配对回放扩展

令 A 表示准备好的 UE5 场景， S 表示归一化动作序列， M 表示角色控制器， θ 表示可编辑物理配置， π_c 表示相机策略。生成轨迹为

$$x = F(A, S, M, \theta, \pi_c). \tag{1}$$

回放扩展固定（A、 S、 M、 π_c），并在不同 θ值下重新运行同一场景。这样，物理编辑成为受控变量，各变体间的差异可归因于编辑后的模拟参数，而非场景内容、输入行为、角色设置或相机规格的变化。

动作序列被视为与场景绑定，因为其语义依赖于局部几何、障碍物和交互目标。因此，流水线将经过验证的场景—动作对，扩展到兼容的物理配置和角色设置上，在保留原始人工设计环境及交互意图的同时生成配对变体。

4.4 同步导出与标注

每次回放通过 Movie Render Queue 与运行时日志导出。Movie Render Queue 生成渲染观测流和辅助渲染通道；运行时日志器记录时间对齐的交互、相机、状态及物理配置元数据。所有导出记录以共享轨迹标识和帧时间线索引，使渲染观测与引擎侧日志能够确定性对齐。

渲染后，流水线执行数据组织、事后语义标注和质量过滤。使用 Qwen3-VL-8B-Instruct [78]为每条轨迹生成片段级描述；描述模型仅观察采样的渲染帧，不接收模拟器元数据。存在渲染失败、同步破坏、严重相机穿模、模拟不稳定，或不能归因于预期物理编辑的发散现象的回放，将被丢弃。

5 Dataset Utility for Gravity-Conditioned Generation

We evaluate PhysEditWorld as supervision for gravity-conditioned generation. Our goal is not to rank generative models comprehensively, but to test whether fine-tuning on matched gravity rollouts improves a model's response to explicit gravity conditions. This distinction matters because visually plausible generation can still fail physically: the camera may remain nearly static, falling may be delayed, or high-gravity rollouts may not accelerate more than low-gravity ones. PhysEditWorld makes these failures measurable by holding the scene, action sequence, camera policy, and initial state fixed while varying gravity.

5.1 Experimental Setup

Data splits. We construct training and evaluation splits at the *replay-group* level rather than the clip level, so that no scene appears in both splits. The training set contains 1,530 (scene, action, gravity) tuples drawn from 9 scenes; the held-out set contains 170 tuples from the remaining 3 scenes. Within each replay group, all gravity variants are kept together to preserve the matched-comparison structure. We sample at most one clip per (scene, action, gravity) triplet to prevent near-duplicate frames from dominating the loss. Gravity multipliers are drawn from $\{0.05, 0.1, 0.5, 1.0, 2.0, 5.0, 20.0\} \times g_{\oplus}$.

Baselines. We evaluate two settings. For gravity-conditioned video generation, we compare zero-shot against **Wan2.2-TI2V-5B** [79] SFT version on our data. Kling 3.0 and Seedance 2.0 appear only in qualitative comparisons as they do not release training code, while Wan2.2-TI2V-5B is additionally fine-tuned via LoRA [80] on PhysEditWorld. For action-conditioned first-person world modeling, we do quality study on **LingBot-World** [5] fine-tuned on our data and compare against the frozen **Matrix-Game 3.0** [6] and **LingBot-World** [5].

Training protocol. We apply LoRA with rank 128 on all attention projection matrices of both backbones, trained for 5 epochs with AdamW (learning rate 1×10^{-4}, batch size 8) on 8×H100 GPUs. Inputs are 5-second clips at 30 FPS, 1280×720, with the gravity multiplier injected as a text token in the conditioning prompt (e.g., `"gravity: 0.25g"`). All other hyperparameters follow each backbone's released SFT recipe.

Evaluation metric. Since generated videos do not expose simulator states or ground-truth trajectories, we recover a per-frame camera trajectory with VGGT [81] and use its vertical-axis component as a one-dimensional fall-progress signal $q(t)$. We restrict evaluation to the pre-landing segment, detected via a simple progress threshold on smoothed $q(t)$, and discard clips with insufficient fall progress or unreliable landing detection. Because VGGT trajectories carry scale ambiguity, all speeds and accelerations are reported in normalized VGGT units, and only relative comparisons within a matched replay group are meaningful. Implementation details (smoothing window, threshold values, rejection rates) are provided.

For each reliable clip, we compute fall-axis speed $v(t)$ as the central-difference derivative of the smoothed $q(t)$ and fit a linear model:

$$v(t) = at + b. \tag{2}$$

The fitted slope a serves as a normalized acceleration proxy, and R^2 measures how well the speed curve follows a linear acceleration pattern. To test whether motion ordering follows the requested gravity ordering, we compute pairwise gravity-acceleration alignment within each matched replay group:

$$\text{Align} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left[(g_i - g_j)(a_i - a_j) > 0\right], \tag{3}$$

where $\mathcal{P} = \{(i,j) \mid i < j,\ i,j \in \text{group}\}$ is the set of unordered pairs within a group, g_i is the requested gravity, and a_i is the fitted acceleration proxy. A perfectly gravity-faithful model achieves $\text{Align} = 1.0$; chance is 0.5.

5.2 Gravity-Conditioned Video Generation / WorldModel

Table 2 reports a representative matched case for Wan2.2-TI2V-5B before and after PhysEditWorld supervised fine-tuning. The zero-shot model is insensitive to the requested gravity: acceleration proxies are near-zero across all three settings, and the alignment of 33.3% confirms that gravity ordering is not preserved. After PhysEditWorld SFT, the acceleration proxy increases monotonically

5 数据集对重力条件生成的作用

我们评估 PhysEditWorld 对重力条件生成的监督作用。目标不是全面评比生成模型，而是检验：利用配对重力轨迹进行微调，是否能改善模型对显式重力条件的响应。这一区别很重要，因为视觉上可信的生成仍可能在物理上失败：相机可能几乎静止，下落可能延迟，或者高重力轨迹的加速度并未超过低重力轨迹。PhysEditWorld 固定场景、动作序列、相机策略和初始状态，只改变重力，因此这些失败可以被量化。

5.1 实验设置

数据划分。我们以回放组而非片段为单位构建训练与评估划分，保证任何场景都不会同时出现在两种划分中。训练集包含来自 9 个场景的 1,530 个（场景、动作、重力）元组；留出集包含其余 3 个场景的 170 个元组。同一回放组中的全部重力变体始终放在一起，以保留配对比较结构。每个（场景、动作、重力）三元组最多采样一个片段，避免近似重复帧主导损失。重力倍数取自 $\{0.05, 0.1, 0.5, 1.0, 2.0, 5.0, 20.0\} \times g_{\oplus}$。

基线。我们评估两种设置。对于重力条件视频生成，将零样本模型与在本数据上监督微调的 Wan2.2-TI2V-5B [79]版本比较。由于 Kling 3.0 和 Seedance 2.0 未公开训练代码，它们仅用于定性比较；Wan2.2-TI2V-5B 则额外通过 LoRA [80]在 PhysEditWorld 上微调。对于动作条件的第一人称世界建模，我们对在本数据上微调的 LingBot-World [5] 开展质量研究，并与冻结的 Matrix-Game 3.**0** [6] 及 LingBot-World [5]比较。

训练协议。对两个骨干网络的所有注意力投影矩阵应用秩为 128 的 LoRA，使用 AdamW 训练 5 个轮次（学习率 1×10^{-4}，批量大小 8），在 8×张 H100 GPU 上运行。输入为 30 FPS、1280×720的 5 秒片段，将重力倍数作为文本标记注入条件提示词（如 "gravity: 0.25g"）。其他超参数遵循各骨干网络公开的监督微调方案。

评估指标。由于生成视频不暴露模拟器状态或真实轨迹，我们用 VGGT [81] 恢复逐帧相机轨迹，并以其竖直轴分量作为一维下落进度信号 $q(t)$。评估仅限着陆前片段，通过平滑后的 $q(t)$ 上的简单进度阈值检测，并丢弃下落进度不足或着陆检测不可靠的片段。由于 VGGT 轨迹具有尺度歧义，所有速度和加速度均以归一化 VGGT 单位报告，只有同一配对回放组内的相对比较有意义。我们提供了平滑窗口、阈值和拒绝率等实现细节。

对每个可靠片段，我们通过平滑后的 $q(t)$ 的中心差分导数，计算下落轴速度 $v(t)$，并拟合线性模型：

$$v(t) = at + b. \tag{2}$$

拟合斜率 a 作为归一化加速度代理量，R^2 衡量速度曲线符合线性加速模式的程度。为检验运动顺序是否遵循指定重力顺序，我们在每个配对回放组内计算重力—加速度的两两对齐率：

$$\text{Align} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left[(g_i - g_j)(a_i - a_j) > 0\right], \tag{3}$$

其中 $\mathcal{P} = \{(i, j) \mid i < j,\ i, j \in \text{group}\}$ 为组内无序对的集合，g_i 为指定重力，a_i 为拟合的加速度代理量。完全忠实于重力的模型达到 $\text{Align} = 1.0$；随机水平为 0.5。

5.2 重力条件视频生成／世界模型

表 2 报告了 Wan2.2-TI2V-5B 在 PhysEditWorld 监督微调前后的一个代表性配对样例。零样本模型对指定重力不敏感：三种设置的加速度代理量均接近零，对齐率 33.3% 证实重力顺序未被保留。PhysEditWorld 监督微调后，加速度代理量单调增大

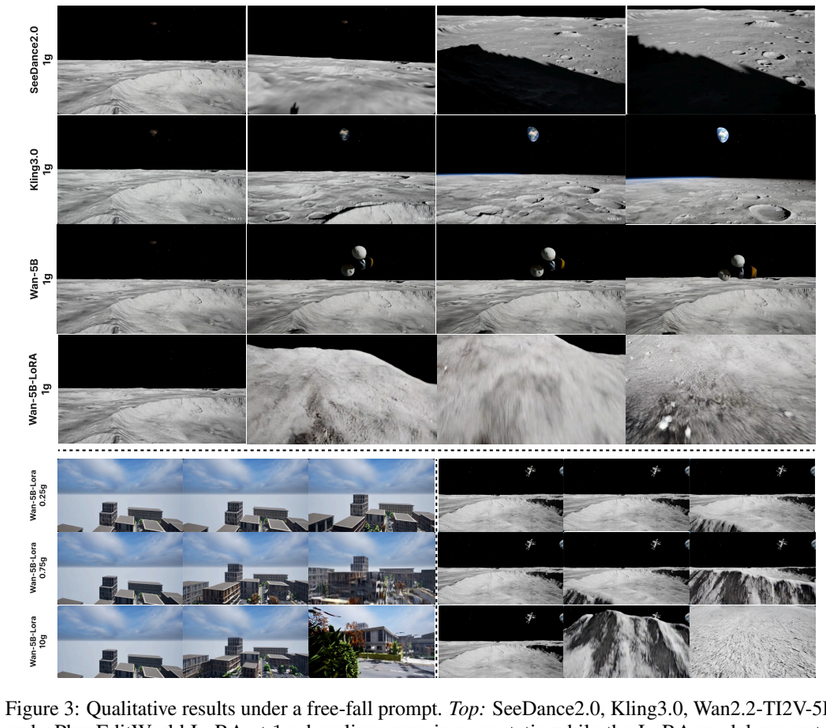

Figure 3: Qualitative results under a free-fall prompt. *Top:* SeeDance2.0, Kling3.0, Wan2.2-TI2V-5B, and +PhysEditWorld LoRA at $1g$; baselines remain near-static while the LoRA model generates strong self-motion with motion blur. *Bottom:* +PhysEditWorld LoRA at $0.25\times$/$0.75\times$/$10\times$ gravity; descent speed scales monotonically, confirming continuously controllable gravity.

with requested gravity, alignment reaches 100%, and mean R^2 rises from 0.066 to 0.570, indicating that the model now responds to gravity as a controllable variable. Figure 3 top shows a qualitative example under a $1g$ prompt: the zero-shot model produces minimal camera motion while the scene remains nearly static, whereas the LoRA-tuned model generates strong forward self-motion toward the lunar surface with visible motion blur.

Table 2: VGGT-based gravity response analysis on a representative held-out matched replay group with three gravity settings ($0.25\times$, $0.75\times$, $10.0\times$). Accel. proxy denotes the fitted slope a in $v(t) = at + b$ on the pre-landing segment, in normalized VGGT units; only *within-row, within-group* comparisons are meaningful. Mean R^2 averages the linear-fit quality across the three settings.

Model	Accel. @ $0.25\times$	Accel. @ $0.75\times$	Accel. @ $10.0\times$	Mean $R^2\uparrow$	Gravity Align. $\uparrow$
Wan2.2-TI2V-5B (zero-shot)	0.0002	0.0046	0.0009	0.066	33.3%
Wan2.2-TI2V-5B + PhysEditWorld SFT	0.1766	0.3983	0.6214	0.570	**100.0%**

5.3 Action-Conditioned First-Person World Modeling

We evaluate whether PhysEditWorld supervision transfers to action-conditioned first-person world models. We use a simple stress test: the `W` key is held continuously from a rooftop edge, where a gravity-faithful model should produce platform departure followed by gravity-dependent downward motion. As shown in Figure 4, both Matrix-Game 3.0 and zero-shot LingBotWorld remain near the platform edge and never enter free-fall. After PhysEditWorld LoRA tuning, LingBotWorld generates platform departure followed by downward self-motion that scales with the requested gravity. This

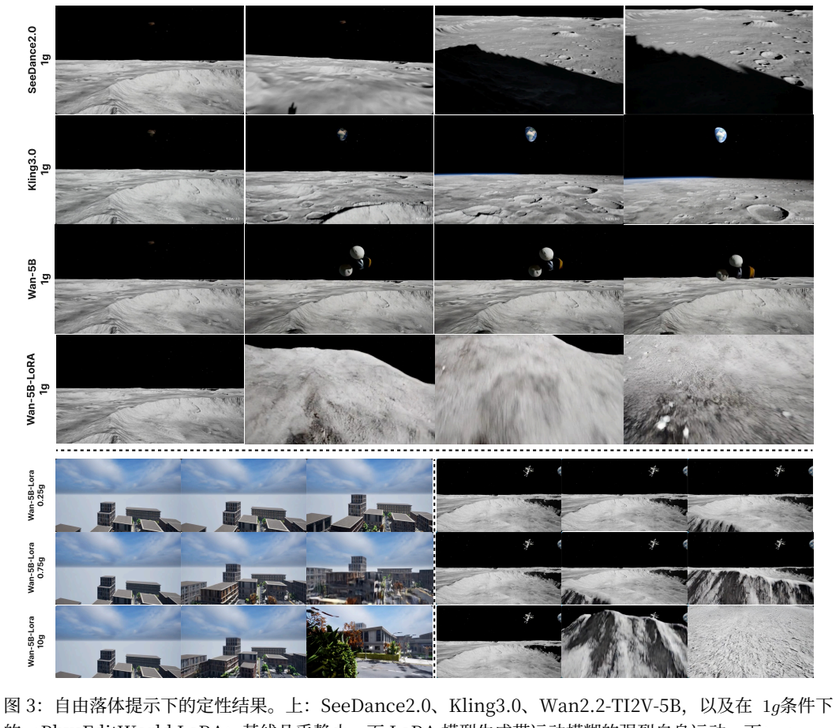

图 3：自由落体提示下的定性结果。上：SeeDance2.0、Kling3.0、Wan2.2-TI2V-5B，以及在 $1g$条件下的 +PhysEditWorld LoRA；基线几乎静止，而 LoRA 模型生成带运动模糊的强烈自身运动。下： +PhysEditWorld LoRA 在 $0.25\times$／$0.75\times$／$10\times$ 重力下的结果；下降速度单调缩放，证实重力可以连续控制。

随指定重力变化，对齐率达到 100%，平均 R^2 从 0.066 升至 0.570，表明模型现在能把重力作为可控变量响应。图 3 上部给出了 $1g$ 提示下的定性示例：零样本模型产生的相机运动极少，场景几乎静止；LoRA 微调模型则生成朝向月球表面的强烈前向自身运动，并伴随可见的运动模糊。

表 2：在一个代表性的留出配对回放组上，基于 VGGT 分析三种重力设置（$0.25\times$、 $0.75\times$、 $10.0\times$）的响应。加速度代理量指着陆前片段中 $v\ (t)\ = at + b$ 的拟合斜率 a ，单位为归一化 VGGT 单位；只有同一行、同一组内的比较有意义。平均 R^2 对三个设置的线性拟合质量取平均。

模型	加速度 @ $0.25\times$	加速度 @ $0.75\times$	加速度 @ $10.0\times$	平均 $R^2\uparrow$	重力对齐率 $\uparrow$
Wan2.2-TI2V-5B（零样本）	0.0002	0.0046	0.0009	0.066	33.3%
Wan2.2-TI2V-5B + PhysEditWorld 监督微调	0.1766	0.3983	0.6214	0.570	**100.0%**

5.3 动作条件第一人称世界建模

我们评估 PhysEditWorld 监督能否迁移到动作条件的第一人称世界模型。采用一个简单的压力测试：从屋顶边缘持续按住 W 键。忠实于重力的模型应先离开平台，随后产生随重力变化的向下运动。如图 4 所示，Matrix-Game 3.0 与零样本 LingBotWorld 均停留在平台边缘附近，始终未进入自由落体。经过 PhysEditWorld LoRA 微调后，LingBotWorld 先生成离开平台，再生成随指定重力缩放的向下自身运动。这一

Figure 4: First-person world-model case study. First frame is generated by GPT-Image2. Baseline models (Matrix-Game 3.0, zero-shot LingBotWorld) remain near the platform edge and never enter free-fall under a $1g$ prompt. After PhysEditWorld LoRA-128 tuning, LingBotWorld generates platform departure and gravity-dependent downward self-motion.

demonstrates that the gravity-response failure is not limited to text-conditioned generation, and that PhysEditWorld provides effective supervision across both generation paradigms.

6 Dataset Utility for Gravity-Aware VLMs

Experimental Setup. We evaluate Qwen3-VL-8B-Instruct [78] on a held-out set of 170 rollouts drawn from the same stratified split described in §5.1. The task requires the model to predict both the gravity class (low / normal / high) and the continuous gravity multiplier from a short gameplay video clip.

Baseline. We first assess the zero-shot capability of Qwen3-VL-8B-Instruct [78] without any domain-specific fine-tuning, serving as a strong off-the-shelf VLM baseline.

Training Protocol. We apply LoRA SFT on the remaining 1,530 training rollouts. To discourage label memorization, gravity targets are jittered by $\pm 10\%$ during training; evaluation is conducted against the original held-out labels.

Metrics. We report class accuracy for the three-way gravity classification, and mean absolute error (MAE), median absolute percentage error (Median APE), and within-10% rate for the continuous gravity multiplier prediction.

Table 3: Gravity-aware prediction on held-out rollouts. +SFT denotes PhysEditWorld LoRA tuning.

Model	Class Acc. (%) ↑	MAE ↓	Median APE (%) ↓	Within 10% ↑
Qwen3-VL-8B[78]	24.71	5.4573	95.00	9.48
Qwen3-VL-8B + SFT	95.29	0.8305	6.22	90.59

As shown in Table 3, the zero-shot model achieves a class accuracy of 24.71%—below the 33.3% random baseline for a three-way classification task—with a median APE of 95.00% and a within-10% rate of only 9.48%, confirming that gravity magnitude is not reliably recoverable from video without targeted supervision. After PhysEditWorld LoRA SFT, class accuracy improves to 95.29% and median APE drops to 6.22%, with 90.59% of predictions falling within 10% of the true gravity

图 4：第一人称世界模型案例研究。首帧由 GPT-Image2 生成。在 $1g$ 提示下，基线模型（Matrix-Game 3.0、零样本 LingBotWorld）停留在平台边缘附近，始终未进入自由落体。经 PhysEditWorld LoRA-128 微调后，LingBotWorld 生成离开平台及随重力变化的向下自身运动。

表明重力响应失效并不限于文本条件生成，PhysEditWorld 能为两种生成范式提供有效监督。

6 数据集对重力感知视觉语言模型的作用

实验设置。我们在 170 条留出轨迹上评估 Qwen3-VL-8B-Instruct [78] ，数据来自第 5.1 节所述的同一分层划分。任务要求模型根据短游戏视频，同时预测重力类别（低／正常／高）和连续重力倍数。

基线。我们首先评估 Qwen3-VL-8B-Instruct [78] 未经任何领域特定微调的零样本能力，将其作为强大的现成视觉语言模型基线。

训练协议。在其余 1,530 条训练轨迹上应用 LoRA 监督微调。为避免标签记忆，训练时对重力目标施加 $\pm 10\%$ 扰动；评估仍使用原始留出标签。

指标。对三分类重力任务报告类别准确率；对连续重力倍数预测报告平均绝对误差（MAE）、绝对百分比误差中位数（Median APE）和误差在 10% 以内的比例。

表 3：留出轨迹上的重力感知预测。 +SFT 表示 PhysEditWorld LoRA 微调。

模型	类别准确率（%）↑	MAE ↓	绝对百分比误差中位数（%）↓	10% 以内 ↑
Qwen3-VL-8B[78]	24.71	5.4573	95.00	9.48
Qwen3-VL-8B + 监督微调	95.29	0.8305	6.22	90.59

如表 3 所示，零样本模型的类别准确率为 24.71%，低于三分类任务 33.3% 的随机基线；绝对百分比误差中位数为 95.00%，误差在 10% 以内的比例仅为 9.48%。这证实，没有针对性监督时，无法可靠地从视频恢复重力大小。经过 PhysEditWorld LoRA 监督微调后，类别准确率提升至 95.29%，绝对百分比误差中位数降至 6.22%，90.59% 的预测落在真实重力的 10% 误差范围内

multiplier. These results demonstrate that the physics-varied rollouts in PhysEditWorld provide effective supervision signal for fine-grained, gravity-aware video-language understanding.

7 Conclusion

PhysEditWorld introduces editable game-world modeling as a controlled physical-intervention problem. Instead of asking only whether a generated rollout looks plausible, it asks whether the same authored scenario evolves consistently when a physical rule is changed. By replaying matched interactions under explicit gravity configurations, PhysEditWorld separates visual realism from physical controllability and reveals failure modes that standard video-quality metrics can overlook. The current release focuses on gravity as a measurable and widely supported first step, covering effects such as airtime, fall speed, jump arcs, and landing dynamics. In future releases, we plan to extend the dataset and pipeline to additional editable physical attributes, such as friction, drag, restitution, wind, and object-level physical parameters, further supporting controllable neural game engines that can be edited as authored simulations rather than only imitated as videos.

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倍数。这些结果表明，PhysEditWorld 中具有物理变化的生成轨迹，为细粒度、重力感知的视频语言理解提供了有效监督信号。

7 结论

PhysEditWorld 将可编辑游戏世界建模定义为受控的物理干预问题。它不只询问生成轨迹是否看起来可信，还询问：改变一条物理规则后，同一人工设计的场景能否一致地演化。通过在显式重力配置下回放配对交互，PhysEditWorld 将视觉逼真度与物理可控性分离，并揭示标准视频质量指标可能忽视的失败模式。当前版本以重力为可测量、广泛支持的第一步，涵盖腾空时长、下落速度、跳跃弧线和着陆动力学等效果。后续版本中，我们计划将数据集和流水线扩展到更多可编辑物理属性，例如摩擦、阻力、恢复系数、风以及物体级物理参数，进一步支持可控神经游戏引擎，使其像人工设计的模拟一样可被编辑，而不仅是模仿视频。

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16

A Asset Library, Character Setup, and Capture Configuration

A.1 Asset Library

PhysEditWorld is built from a manually curated UE5 asset library designed to cover diverse gravity-sensitive interaction scenarios. The current asset collection contains 12 representative scene environments, including polar research bases, indoor sports courts, lunar and Martian surfaces, cyberpunk city blocks, classrooms, laboratories, forest valleys, research camps, bowling alleys, theaters, and modern urban streets. These assets span indoor, outdoor, planetary, urban, sports, and object-interaction settings, allowing the dataset to expose different forms of gravity-dependent behavior such as free fall, jump arcs, landing timing, object displacement, and contact response. Artists manually inspect each scene for visual fidelity, collision quality, interaction suitability, and physical stability before it is included in the replay pipeline.

In addition to scene assets, PhysEditWorld includes three character assets with compatible animation and retargeting setups. The character system uses a UE5 animation stack based on motion matching, pose search, chooser-driven animation selection, blend stacks, orientation warping, and animation warping. To support low-gravity scenarios, we implement a low-gravity animation tuner that adjusts airborne-state detection and animation playback rate according to the gravity multiplier. This prevents low-gravity clips from using visually implausible Earth-gravity animation timing. The camera system supports socket-based attachment to skeletal bones, such as head, spine, or pelvis sockets, so first-person and actor-following views can inherit smooth character motion during jumping, falling, and landing. External character meshes are connected to the animation library through IK Rig and IK Retargeter assets, enabling different characters to share a consistent motion-control interface during counterfactual replay.

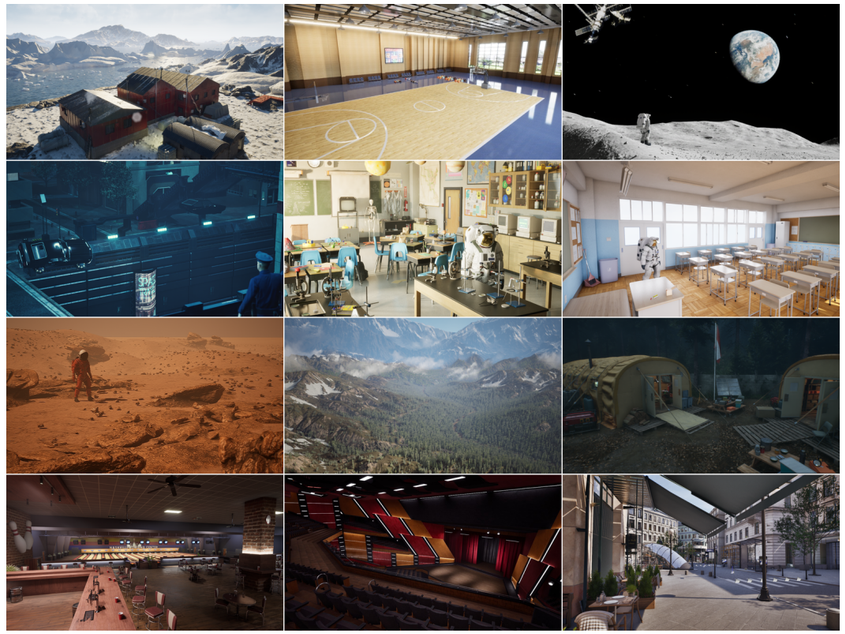

Figure 5: The 12 curated UE5 scene assets used in PhysEditWorld. The asset library covers indoor, outdoor, planetary, urban, sports, and object-interaction environments to support diverse gravity-sensitive motion patterns.

A 资产库、角色设置与采集配置

A.1 资产库

PhysEditWorld 基于人工筛选的 UE5 资产库构建，旨在覆盖多样的重力敏感交互场景。当前资产集合包含 12 个代表性场景环境，包括极地研究基地、室内运动场、月球和火星表面、赛博朋克城市街区、教室、实验室、森林山谷、研究营地、保龄球馆、剧院以及现代城市街道。这些资产涵盖室内、室外、行星、城市、运动与物体交互场景，使数据集呈现自由落体、跳跃弧线、着陆时序、物体位移和接触响应等不同类型的重力相关行为。每个场景进入回放流水线前，艺术人员都会人工检查其视觉保真度、碰撞质量、交互适用性和物理稳定性。

除场景资产外，PhysEditWorld 还包含三个角色资产，并配有兼容的动画与重定向设置。角色系统采用基于运动匹配、姿态搜索、选择器驱动的动画选择、混合栈、朝向扭曲和动画扭曲的 UE5 动画栈。为支持低重力场景，我们实现了低重力动画调节器，根据重力倍数调整腾空状态检测与动画播放速率，从而避免低重力视频沿用地球重力下不合理的动画时序。相机系统支持连接到骨骼插槽，如头部、脊柱或骨盆插槽，使第一人称及跟随角色的视角在跳跃、下落和着陆时继承平滑的角色运动。外部角色网格通过 IK Rig 与 IK Retargeter 资产接入动画库，使不同角色在反事实回放时共享一致的运动控制接口。

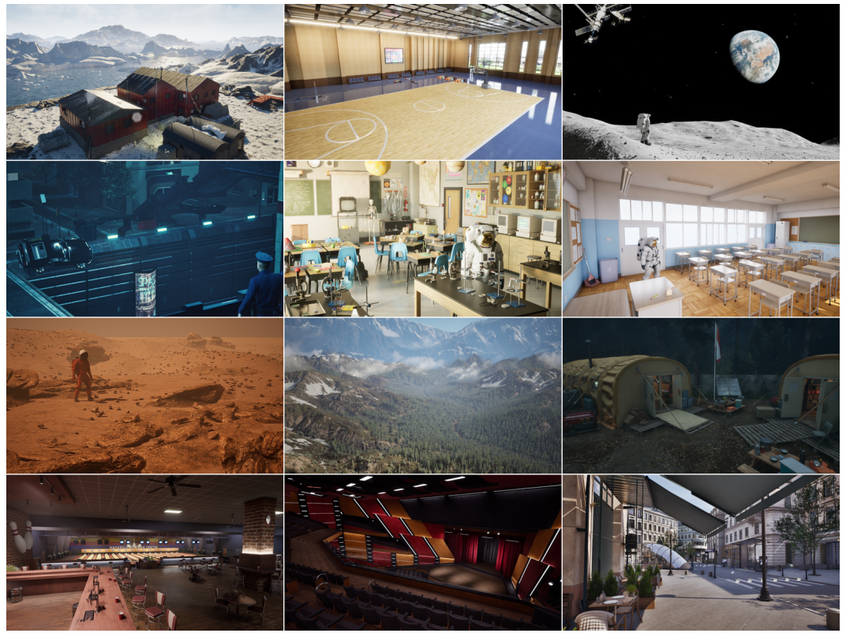

图 5：PhysEditWorld 使用的 12 个精选 UE5 场景资产。资产库涵盖室内、室外、行星、城市、运动和物体交互环境，以支持多样的重力敏感运动模式。

Figure 6: Character assets used in the current PhysEditWorld pipeline. Characters are configured with compatible controllers, animation retargeting, and low-gravity animation tuning to support reproducible replay under edited gravity settings.

PhysEditWorld is built from a manually curated UE5 asset library designed to cover diverse gravity-sensitive interaction scenarios. The current asset collection contains 12 representative scene environments, including polar research bases, indoor sports courts, lunar and Martian surfaces, cyberpunk city blocks, classrooms, laboratories, forest valleys, research camps, bowling alleys, theaters, and modern urban streets. These assets span indoor, outdoor, planetary, urban, sports, and object-interaction settings, allowing the dataset to expose different forms of gravity-dependent behavior such as free fall, jump arcs, landing timing, object displacement, and contact response. Artists manually inspect each scene for visual fidelity, collision quality, interaction suitability, and physical stability before it is included in the replay pipeline.

In addition to scene assets, PhysEditWorld includes character assets with compatible controller, animation, and retargeting setups. The character system uses a UE5 animation stack based on motion matching, pose search, chooser-driven animation selection, blend stacks, orientation warping, and animation warping. To support low-gravity scenarios, we implement a low-gravity animation tuner that adjusts airborne-state detection and animation playback rate according to the gravity multiplier. This prevents low-gravity clips from using visually implausible Earth-gravity animation timing. External character meshes are connected to the shared animation library through IK Rig and IK Retargeter assets, enabling different characters to share a consistent motion-control interface during counterfactual replay.

A.2 Character-Centric Multi-Camera Capture

The capture system uses a character-centric multi-camera rig rather than placing independent camera actors in the scene. Multiple `UCameraComponent`s are embedded inside the character blueprint and attached through skeletal sockets or bones, spring arms, and camera components. This topology provides a shared character-centered reference frame for all camera views, so first-person, third-person, side, front, back, and oblique views remain temporally aligned during walking, turning, jumping, falling, and landing. In the current setup, the camera array includes `FP`, `BK`, `FL`, `FW`, `FR`, `LF`, `RT`, and `TP` views. Rendering all views from the same replay instance avoids the synchronization drift that would arise from repeated simulation or separately placed cameras.

Socket-based attachment also supports stable close-range observation. Cameras can be mounted to head, spine, pelvis, or other skeletal sockets, while spring arms decouple the choice of skeletal anchor from the camera offset and near-body framing. During interactive preview, spring arms can provide collision handling and visual smoothing. During dataset replay and rendering, camera lag is disabled to prioritize deterministic frame alignment and reproducible multi-view sampling. This design is especially important for gravity-edited rollouts, where jump and fall timing are the measurement target rather than merely cinematic motion.

图 6：当前 PhysEditWorld 流水线使用的角色资产。角色配有兼容控制器、动画重定向及低重力动画调节，以支持编辑重力设置下的可复现回放。

PhysEditWorld 基于人工筛选的 UE5 资产库构建，旨在覆盖多样的重力敏感交互场景。当前资产集合包含 12 个代表性场景环境，包括极地研究基地、室内运动场、月球和火星表面、赛博朋克城市街区、教室、实验室、森林山谷、研究营地、保龄球馆、剧院以及现代城市街道。这些资产涵盖室内、室外、行星、城市、运动与物体交互场景，使数据集呈现自由落体、跳跃弧线、着陆时序、物体位移和接触响应等不同类型的重力相关行为。每个场景进入回放流水线前，艺术人员都会人工检查其视觉保真度、碰撞质量、交互适用性和物理稳定性。

除场景资产外，PhysEditWorld 还包含角色资产，以及兼容的控制器、动画和重定向设置。角色系统采用基于运动匹配、姿态搜索、选择器驱动的动画选择、混合栈、朝向扭曲和动画扭曲的 UE5 动画栈。为支持低重力场景，我们实现了低重力动画调节器，根据重力倍数调整腾空状态检测与动画播放速率，避免低重力视频采用视觉上不合理的地球重力动画时序。外部角色网格通过 IK Rig 和 IK Retargeter 资产接入共享动画库，使不同角色在反事实回放期间共享一致的运动控制接口。

A.2 角色中心多相机采集

采集系统采用以角色为中心的多相机装置，而非在场景中独立放置相机实体。多个 UCameraComponent 嵌入角色蓝图，并通过骨骼插槽或骨骼、弹簧臂及相机组件连接。这种拓扑为所有相机视角提供共享的角色中心参考系，使第一人称、第三人称、侧面、前方、后方和斜向视角在行走、转向、跳跃、下落和着陆时保持时间对齐。当前设置的相机阵列包含 FP、BK、FL、FW、FR、LF、RT 和 TP 视角。由同一次回放实例渲染全部视角，可避免重复模拟或分别放置相机带来的同步漂移。

基于插槽的连接也支持稳定的近距离观察。相机可安装到头部、脊柱、骨盆或其他骨骼插槽，而弹簧臂将骨骼锚点的选择与相机偏移、近身构图解耦。交互预览时，弹簧臂可提供碰撞处理与视觉平滑；数据集回放和渲染时则关闭相机延迟，优先保证确定性的逐帧对齐与可复现的多视角采样。这一设计对重力编辑轨迹尤为重要，因为跳跃和下落时序是测量目标，而不只是电影化运动。

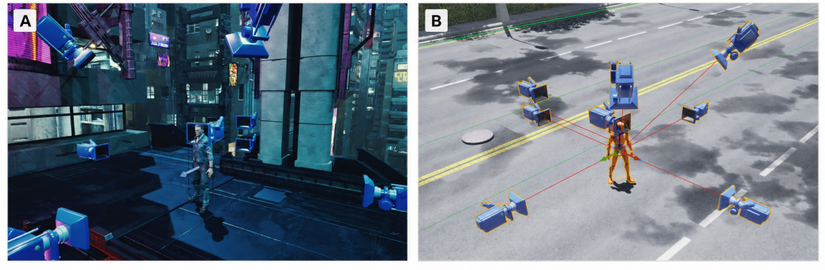

Figure 7: Character-centric multi-camera rig used for synchronized capture. (A) Cameras are embedded around the actor inside a UE5 scene. (B) The rig defines multiple named views around the same character reference frame, including first-person, third-person, side, front, and back views.

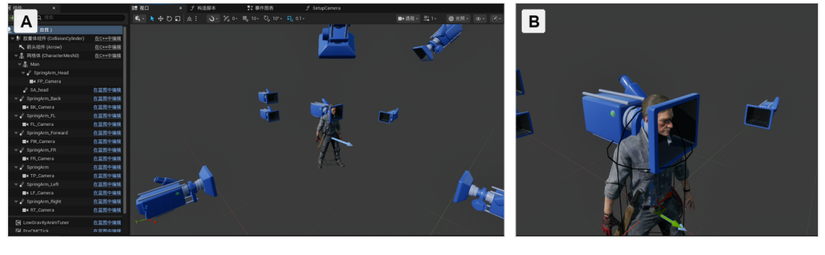

Figure 8: Socket and spring-arm based camera setup. (A) The character blueprint contains multiple spring-arm and camera components attached around the skeletal mesh. (B) Close-range cameras are mounted through skeletal sockets and spring arms, allowing first-person or near-body views to inherit character motion while preserving controllable camera offsets.

A.3 Render Graph Outputs and Metadata

The UE5 rendering job is configured as a multi-branch Movie Render Graph. A shared global settings chain controls warm-up, game overrides, camera settings, and output settings. The current configuration uses a warm-up interval before formal capture so that animation state, temporal antialiasing, post-processing, and motion blur reach a stable state before frames are written. Captures are exported at 1280×720 resolution and 30 FPS.

The render graph produces several synchronized output branches from the same replay. An object-ID or segmentation branch exports per-frame ID supervision. An 8-bit image branch exports standard image sequences and auxiliary render passes such as motion vectors and world normals. A video branch exports H.264 MP4 previews with audio for manual inspection and semantic quality control. A 16-bit PNG branch exports high-precision depth-style supervision. In parallel, replay metadata is written as frame-level and time-level CSV files, together with a camera-name mapping JSON file. These metadata records include replay input events, player and camera transforms, physical state, camera identity, and capture termination signals.

B UE Editor Plugin and Data Generation Workflow

PhysEditWorld is generated through a UE5 Editor plugin that manages the data-generation workflow from scene preparation to large-scale synthetic data production. The plugin is designed to operate directly on artist-authored UE5 levels with minimal intrusion into existing game content. Instead

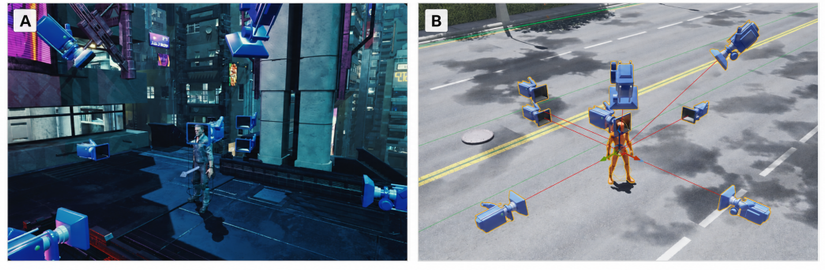

图 7：用于同步采集的角色中心多相机装置。（A）相机在 UE5 场景中嵌入角色周围。（B）装置围绕同一角色参考系定义多个命名视角，包括第一人称、第三人称、侧面、前方与后方。

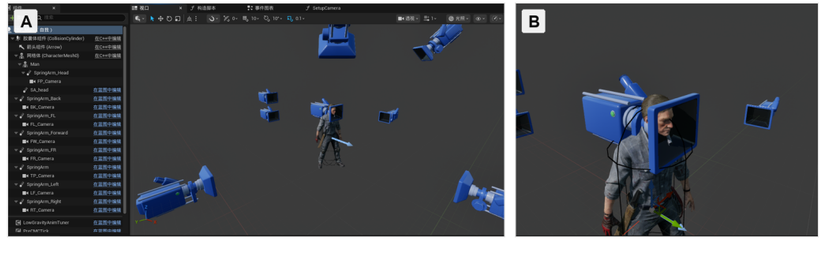

图 8：基于插槽和弹簧臂的相机设置。（A）角色蓝图包含围绕骨骼网格连接的多个弹簧臂与相机组件。（B）近距离相机通过骨骼插槽和弹簧臂安装，使第一人称或近身视角继承角色运动，同时保留可控的相机偏移。

A.3 渲染图输出与元数据

UE5 渲染任务配置为多分支 Movie Render Graph。共享的全局设置链控制预热、游戏覆写、相机设置和输出设置。当前配置在正式采集前设有预热时段，使动画状态、时间抗锯齿、后处理及运动模糊在写入帧前达到稳定状态。输出采用 1280×720 分辨率与 30 FPS。

渲染图从同一次回放产生多个同步输出分支。物体 ID 或分割分支导出逐帧 ID 监督；8 位图像分支导出标准图像序列，以及运动向量、世界法线等辅助渲染通道；视频分支导出带音频的 H.264 MP4 预览，供人工检查与语义质量控制；16 位 PNG 分支导出高精度的深度类监督。同时，回放元数据写入帧级与时间级 CSV 文件，并附带相机名称映射 JSON 文件。这些元数据记录包括回放输入事件、玩家与相机变换、物理状态、相机身份和采集终止信号。

B UE 编辑器插件与数据生成流程

PhysEditWorld 通过 UE5 编辑器插件生成，该插件管理从场景准备到大规模合成数据生产的数据生成流程。插件以对现有游戏内容的最小侵入，直接处理艺术人员设计的 UE5 关卡。它并非

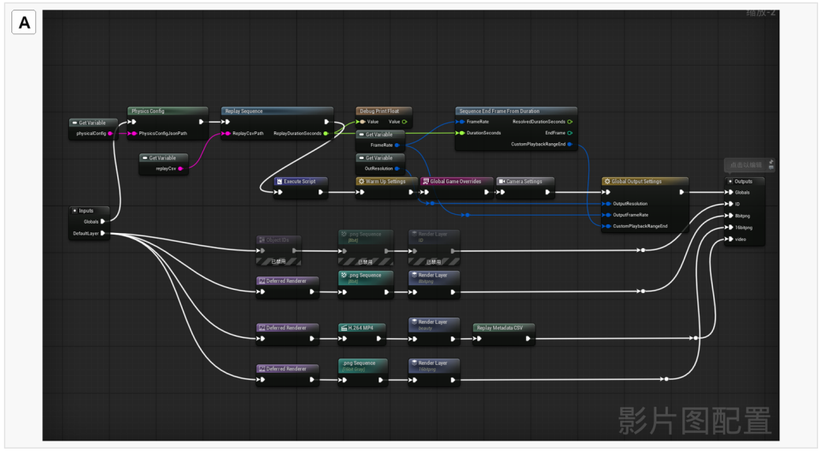

Figure 9: Movie Render Graph configuration for synchronized multimodal export. The graph combines replay sequence loading, warm-up, global game and camera settings, output settings, multi-branch visual exports, H.264 preview video, high-precision depth output, and replay metadata CSV export.

of rebuilding scenes in a separate simulator, it augments prepared levels with the runtime contexts, replay components, camera bindings, and batch-generation interfaces required for reproducible replay-and-rendering.

A key design goal of the plugin is to make scene preparation lightweight. As illustrated in Figure ??, each artist-prepared level can be converted into a production-ready data-generation scene through four editor operations: selecting the required EnhancedInputContext, specifying the sequence and camera bindings, running the FactoryManager initialization, and saving the produce-ready level. Since most updated character controller implementations already expose Enhanced Input support, this design remains extensible across different controllers and input schemes.

After the level-specific configuration is completed, the FactoryManager initializes the scene with a single command. It automatically registers the data-collection context, attaches replay components, inserts the required capture and rendering modules, and validates the camera and input bindings. This step injects only the minimal components needed for data generation, preserving the original authored scene and gameplay logic as much as possible. Once initialization finishes, the level becomes ready for reproducible replay and rendering.

The plugin then supports two ways of constructing interaction sequences. The first is Play-in-Editor interaction capture, where users directly play the prepared level and record semantic input-action traces, including movement axes, jump commands, camera deltas, and button states. The second is procedural sequence generation, where PCG-based scripts generate scalable interaction sequences for broader coverage. We also combined the Navmesh Agent in our pipline. to Both sources are converted into replayable interaction sequences that can be executed under controlled physical configurations.

For large-scale production, the DataFactory pipeline exposes Python interfaces and YAML configuration files for composing replay, rendering, cleaning, merging, and recovery stages. The batch system supports checkpointing and resume functionality, allowing interrupted jobs to continue from the last valid state rather than restarting from scratch. This makes the same editor-plugin workflow usable both for interactive debugging inside UE5 and for large-scale offline dataset generation.

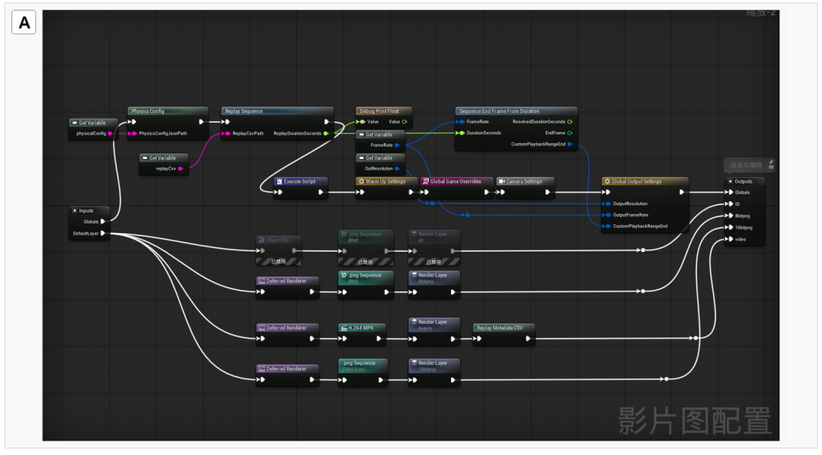

图 9：用于同步多模态导出的 Movie Render Graph 配置。该图结合回放序列加载、预热、全局游戏及相机设置、输出设置、多分支视觉导出、H.264 预览视频、高精度深度输出和回放元数据 CSV 导出。

在独立模拟器中重建场景，而是为准备好的关卡添加可复现回放与渲染所需的运行时上下文、回放组件、相机绑定及批量生成接口。

插件的一项关键设计目标是简化场景准备。如图 ?? 所示，每个艺术人员准备的关卡可通过四项编辑器操作转化为可投入生产的数据生成场景：选择所需 EnhancedInputContext，指定序列与相机绑定，运行 FactoryManager 初始化，保存可投入生产的关卡。由于多数更新后的角色控制器实现已暴露 Enhanced Input 支持，该设计可扩展至不同控制器与输入方案。

关卡特定配置完成后，FactoryManager 用一条命令初始化场景。它自动注册数据采集上下文、连接回放组件、插入必要采集与渲染模块，并验证相机和输入绑定。该步骤仅注入数据生成所需的最少组件，尽可能保留原始人工设计的场景和游戏逻辑。初始化完成后，关卡即可进行可复现回放与渲染。

插件随后支持两种构建交互序列的方式。第一种是编辑器内运行的交互采集：用户直接游玩准备好的关卡，记录语义输入动作轨迹，包括移动轴、跳跃命令、相机增量和按钮状态。第二种是程序化序列生成：基于 PCG 的脚本生成可扩展的交互序列，以获得更广覆盖。我们还在流水线中集成了 NavmeshAgent。两类来源均被转换为可回放的交互序列，在受控物理配置下执行。

为支持大规模生产，DataFactory 流水线提供 Python 接口与 YAML 配置文件，用于组合回放、渲染、清洗、合并和恢复阶段。批处理系统支持检查点与恢复，使中断任务可从最后有效状态继续，无需从头重启。因此，同一编辑器插件流程既可用于 UE5 内的交互式调试，也可用于大规模离线数据集生成。

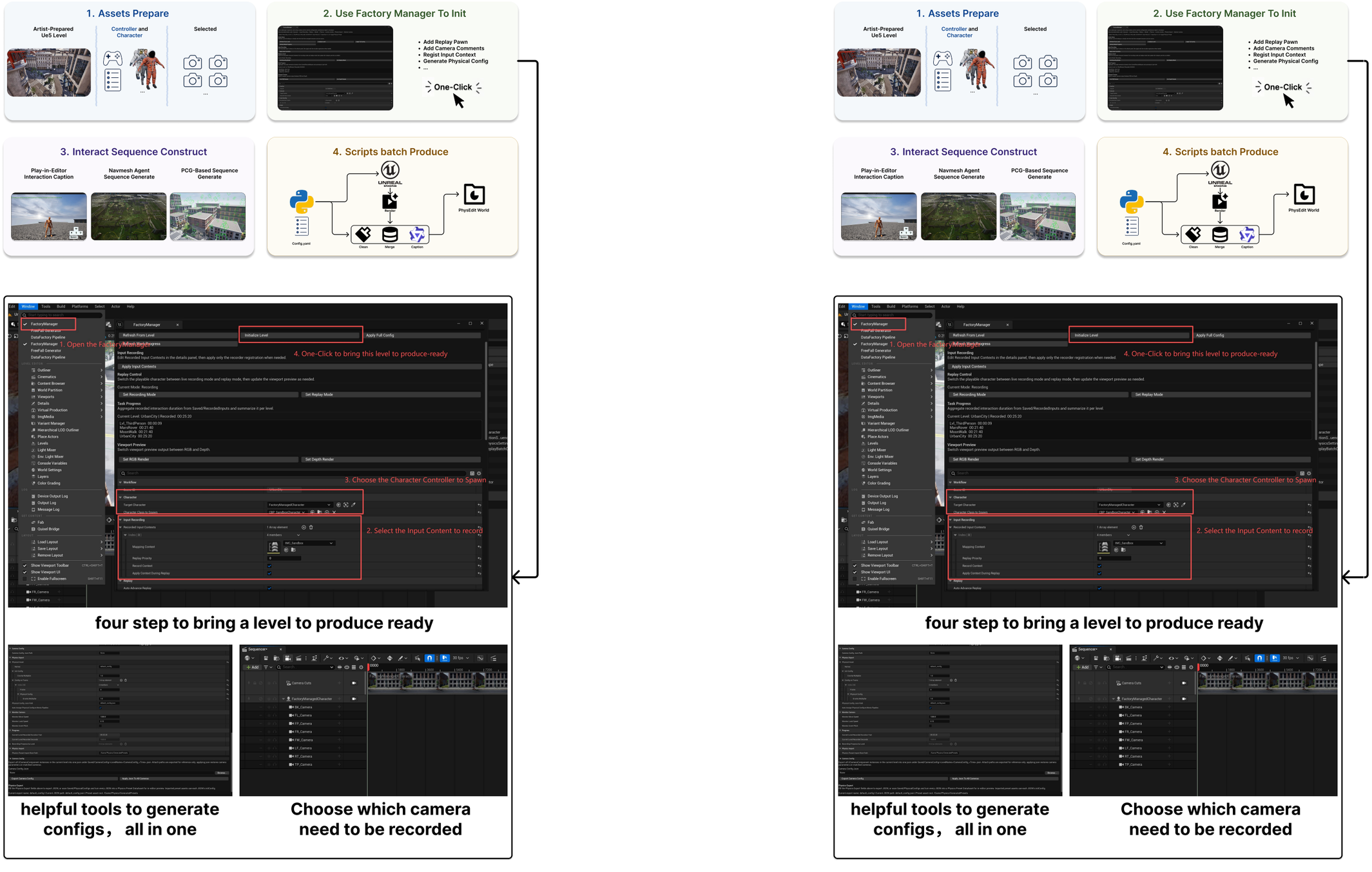

Figure 10: UE Editor Plugin and Data Generation Workflow

1. Assets Prepare
Artist-Prepared Ue5 Level
Controller and Character
Selected
2. Use Factory Manager To Init
Add Replay Pawn
Add Camera Comments
Regist Input Context
Generate Physical Config
...
One-Click
3. Interact Sequence Construct
Play-in-Editor Interaction Caption
Navmesh Agent Sequence Generate
PCG-Based Sequence Generate
4. Scripts batch Produce
UNREAL
Render
Config.yaml
Clean
Merge
Caption
PhysEdit World
1. Open the FacaryMnager Window
4. One-Click to bring this level to produce-ready
3. Choose the Character Controller to Spawn
2. Select the Input Content to record
four step to bring a level to produce ready
helpful tools to generate configs， all in one
Choose which camera need to be recorded

图 10：UE 编辑器插件与数据生成流程

21

C Dataset Sample Format and Metadata Schema

Each data sample in our dataset corresponds to one complete recording-rendering job under a fixed scene, action sequence, camera setup, and physical configuration. We use an 8-digit string as the sample identifier, e.g., `00000000`. A sample is self-contained and consists of synchronized multi-view RGB videos, frame-level metadata, event-level metadata, and a physical-configuration descriptor. The dataset root contains a global index file, `Details.json`, which stores the list of samples and the relative paths to their associated files.

The cleaned dataset follows the directory layout below:

```
<dataset_root>/
  Video/<sample_id>/<camera_name>.mp4
  Meta/<sample_id>_meta_frame.csv
  Meta/<sample_id>_meta_time.csv
  PhysicalConfig/<physical_config_name>.json
  Details.json
```

For auxiliary modalities, such as depth, surface normals, or object masks, the optional frame-wise outputs are stored as:

`Aux/<aux_type>/<sample_id>/<camera_name>/<frame_id>.png`

Table 4 summarizes the main fields used to describe each sample.

Table 4: Main fields of a dataset sample.

Field	Description
`sample_id`	Unique 8-digit identifier of the sample.
`scene_name`	Name of the simulated scene or environment.
`action_sequence_name`	Identifier of the replayed action or motion sequence.
`physical_config_name`	Name of the physical condition used for simulation.
`physical_config_file`	Relative path to the physical-configuration JSON file.
`frame_meta_file`	Relative path to frame-level metadata.
`time_meta_file`	Relative path to event-level temporal metadata.
`camera_names`	List of camera names available in the sample.
`cameras`	Per-camera video paths and attributes.
`aux_types`	Optional auxiliary modalities associated with the sample.

A representative sample entry is shown below. All paths are relative to the dataset root.

```
{
  "sample_id": "00000000",
  "scene_name": "Map_MarsRover",
  "action_sequence_name": "MarsSequence_0001",
  "physical_config_name": "G_0.05",
  "physical_config_file": "PhysicalConfig/G_0.05.json",
  "frame_meta_file": "Meta/00000000_meta_frame.csv",
  "time_meta_file": "Meta/00000000_meta_time.csv",
  "camera_names": ["BK_Camera", "FL_Camera", "FP_Camera", ... ],
  "aux_types": ["Depth", "Normals"],
  "cameras": [
    {
      "camera_name": "BK_Camera",
      "rgb_file": "Video/00000000/BK_Camera.mp4",
      "aux": {
        "Depth": "Aux/Depth/00000000/BK_Camera",
        "Normals": "Aux/Normals/00000000/BK_Camera",
        "ObjectMask": "Aux/ObjectMask/00000000/BK_Camera"
      },
```


C 数据集样本格式与元数据模式

数据集中的每个样本对应固定场景、动作序列、相机设置和物理配置下的一次完整录制—渲染任务。样本标识符为 8 位字符串，如 00000000。样本自包含，由同步多视角 RGB 视频、帧级元数据、事件级元数据和物理配置描述符构成。数据集根目录包含全局索引文件 Details.json，存储样本列表及关联文件的相对路径。

清洗后的数据集采用以下目录布局：

```
<dataset_root>/
  Video/<sample_id>/<camera_name>.mp4
  Meta/<sample_id>_meta_frame.csv
  Meta/<sample_id>_meta_time.csv
  PhysicalConfig/<physical_config_name>.json
  Details.json
```

深度、表面法线、物体掩码等可选辅助模态的逐帧输出存储为：

Aux/<aux_type>/<sample_id>/<camera_name>/<frame_id>.png

表 4 概述了描述每个样本的主要字段。

表 4：数据集样本的主要字段。

字段	说明
`sample_id`	样本的唯一 8 位标识符。
`scene_name`	模拟场景或环境的名称。
`action_sequence_name`	回放动作或运动序列的标识符。
`physical_config_name`	模拟所用物理条件的名称。
`physical_config_file`	物理配置 JSON 文件的相对路径。
`frame_meta_file`	帧级元数据的相对路径。
`time_meta_file`	事件级时间元数据的相对路径。
`camera_names`	样本中可用相机名称的列表。
`cameras`	各相机的视频路径与属性。
`aux_types`	样本关联的可选辅助模态。

下面展示一个代表性样本条目。所有路径均相对于数据集根目录。

```
{
  "sample_id": "00000000",
  "scene_name": "Map_MarsRover",
  "action_sequence_name": "MarsSequence_0001",
  "physical_config_name": "G_0.05",
  "physical_config_file": "PhysicalConfig/G_0.05.json",
  "frame_meta_file": "Meta/00000000_meta_frame.csv",
  "time_meta_file": "Meta/00000000_meta_time.csv",
  "camera_names": ["BK_Camera", "FL_Camera", "FP_Camera", ... ],
  "aux_types": ["Depth", "Normals"],
  "cameras": [
    {
      "camera_name": "BK_Camera",
      "rgb_file": "Video/00000000/BK_Camera.mp4",
      "aux": {
        "Depth": "Aux/Depth/00000000/BK_Camera",
        "Normals": "Aux/Normals/00000000/BK_Camera",
        "ObjectMask": "Aux/ObjectMask/00000000/BK_Camera"
      },
```

```
      "resolution": {"width": 1280, "height": 720},
      "fps": 30
    },
    ...
  ]
}
```

The two metadata files provide complementary temporal annotations. The frame-level file records information indexed by rendered frame, which is used to align visual observations with simulation states. The event-level file records simulation and capture events over time, such as the start and end of recording or replay. The physical-configuration file stores the controlled simulation parameters, enabling downstream methods to condition learning or evaluation on explicit physical settings.

D Example usage.

The dataset generation pipeline is driven by a YAML configuration file and is launched through the repository-level entry point `Scripts/cli.py`. The configuration specifies the simulated scenes, replay trajectories, physical-configuration files, Movie Render Graph preset, output resolution, frame rate, and the enabled pipeline stages.

Before running the full pipeline, we first validate the configuration and inspect the planned jobs:

```
uv run --with pyyaml python Scripts/cli.py \
  --config Scripts/configs/base.yaml \
  --dry-run
```

After verifying the generated plan, we run the complete rendering and cleaning pipeline:

```
uv run --with pyyaml python Scripts/cli.py \
  --config Scripts/configs/base.yaml
```

A typical configuration contains the following fields:

```
base_path: <project_root>
unreal_exe: <path_to_UnrealEditor>
dataset_path: <raw_dataset_root>
output_path: <cleaned_dataset_root>
input_sequence: <replay_csv_root>
physical_config: <physical_config_root>

scenes:
  - name: <scene_name>
    level_path: <unreal_level_asset_path>
    level_sequence_path: <unreal_level_sequence_asset_path>

job:
  selection: auto
  mrg_path: <movie_render_graph_asset_path>
  params:
    fps: 30
    resolution: [1280, 720]
    mp4: true

pipeline:
  render:
    enable: true
    auto_start: true
  clean:
    enable: true
    input_root: <stage1_render_output_root>
```



```
      "resolution": {"width": 1280, "height": 720},
      "fps": 30
    },
    ...
  ]
}
```


两个元数据文件提供互补的时间标注。帧级文件记录按渲染帧索引的信息，用于对齐视觉观测与模拟状态；事件级文件记录随时间发生的模拟和采集事件，如录制或回放的开始与结束。物理配置文件保存受控模拟参数，使下游方法能够以显式物理设置为条件开展学习或评估。

D 使用示例

数据集生成流水线由 YAML 配置文件驱动，通过仓库级入口 Scripts/cli.py 启动。配置指定模拟场景、回放轨迹、物理配置文件、Movie Render Graph 预设、输出分辨率、帧率及启用的流水线阶段。

运行完整流水线前，先验证配置并检查计划任务


```
uv run --with pyyaml python Scripts/cli.py \
  --config Scripts/configs/base.yaml \
  --dry-run
```


验证生成计划后，运行完整渲染与清洗流水线：


```
uv run --with pyyaml python Scripts/cli.py \
  --config Scripts/configs/base.yaml
```


典型配置包含以下字段：


```
base_path: <project_root>
unreal_exe: <path_to_UnrealEditor>
dataset_path: <raw_dataset_root>
output_path: <cleaned_dataset_root>
input_sequence: <replay_csv_root>
physical_config: <physical_config_root>

scenes:
  - name: <scene_name>
    level_path: <unreal_level_asset_path>
    level_sequence_path: <unreal_level_sequence_asset_path>

job:
  selection: auto
  mrg_path: <movie_render_graph_asset_path>
  params:
    fps: 30
    resolution: [1280, 720]
    mp4: true

pipeline:
  render:
    enable: true
    auto_start: true
  clean:
    enable: true
    input_root: <stage1_render_output_root>
```

```
output_root: <cleaned_dataset_root>
physical_config_root: <physical_config_root>
append: false
workers: 8
overwrite: false
dataset_name: <dataset_name>
```

The command above expands the configuration into rendering jobs over the Cartesian product of scenes, replay trajectories, and physical configurations. It then launches Unreal Engine to execute the rendering jobs and, after rendering, converts the stage-1 outputs into the canonical dataset layout described in Appendix C.

When the stage-1 rendering outputs have already been generated, the cleaning step can also be executed independently:

```
uv run python Scripts/utools/clean_mrq_graph_dataset.py \
  <stage1_render_output_root> \
  --output-root <cleaned_dataset_root> \
  --physical-config-root <physical_config_root> \
  --workers 8 \
  --dataset-name <dataset_name>
```

This cleaning-only command scans the stage-1 output directories, identifies valid rendering jobs by the presence of `camera_name_map.json` and camera videos, copies the RGB videos and metadata into the canonical layout, and writes the global index file `Details.json`.

E More Asset

A small set of data is showed in the supplementary material with a html for the limitation of size.

F Release Plan

We plan to publicly release the PhysEditWorld dataset, including synchronized RGB videos, depth maps, normal maps, gravity annotations, camera trajectories, action traces, and evaluation scripts used in this work. We have provided a small subset in out supplementary material for the size limitation.

We also plan to release the UE5-based replay-and-rendering pipeline **as a plugin**, together with data generation configurations and example replay assets, to facilitate reproducibility and future research on physics-editable world modeling.Due to potential licensing restrictions associated with certain third-party UE5 assets, some raw scene assets or commercial content may not be redistributed directly. In such cases, we will provide replacement assets, asset lists, or reconstruction instructions whenever possible.


```
output_root: <cleaned_dataset_root>
physical_config_root: <physical_config_root>
append: false
workers: 8
overwrite: false
dataset_name: <dataset_name>
```

上述命令将配置展开为场景、回放轨迹和物理配置笛卡尔积上的渲染任务。随后启动 UnrealEngine 执行渲染，并在完成后将第一阶段输出转换为附录 C 所述的规范数据集布局。

若第一阶段渲染输出已生成，也可独立执行清洗步骤：

```
uv run python Scripts/utools/clean_mrq_graph_dataset.py \
  <stage1_render_output_root> \
  --output-root <cleaned_dataset_root> \
  --physical-config-root <physical_config_root> \
  --workers 8 \
  --dataset-name <dataset_name>
```

此仅清洗命令扫描第一阶段输出目录，根据 camera_name_map.json 和相机视频的存在识别有效渲染任务，将 RGB 视频与元数据复制到规范布局，并写入全局索引文件 Details.json。

E 更多资产

由于大小限制，补充材料通过 HTML 展示一小部分数据。

F 发布计划

我们计划公开发布 PhysEditWorld 数据集，包括同步 RGB 视频、深度图、法线图、重力标注、相机轨迹、动作轨迹，以及本文使用的评估脚本。受大小限制，我们已在补充材料中提供一个小型子集。

我们还计划以插件形式发布基于 UE5 的回放与渲染流水线，同时提供数据生成配置与示例回放资产，以促进可复现性和物理可编辑世界建模的后续研究。由于部分第三方 UE5 资产可能有许可限制，一些原始场景资产或商业内容可能无法直接再分发。此时，我们将尽可能提供替代资产、资产清单或重建说明。

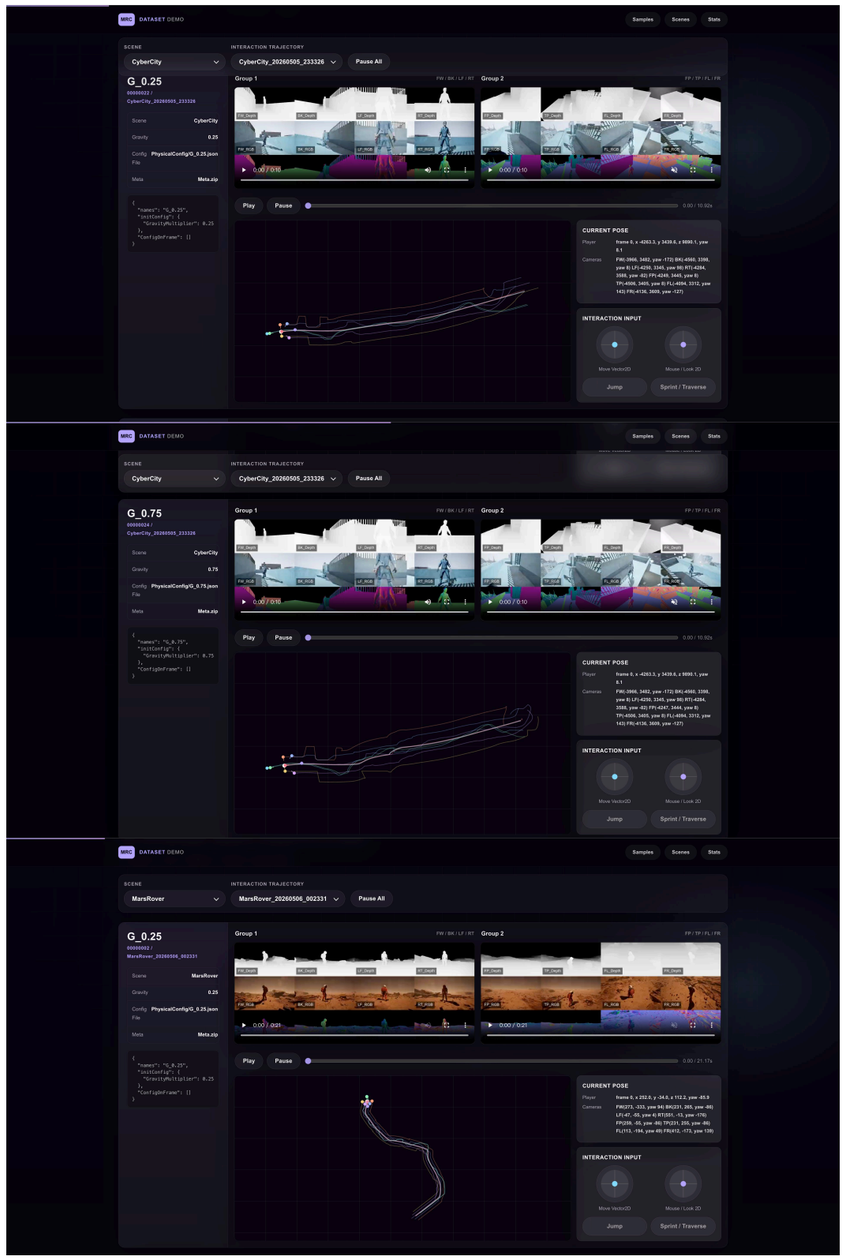

Figure 11: More PhysicsEdit-Data

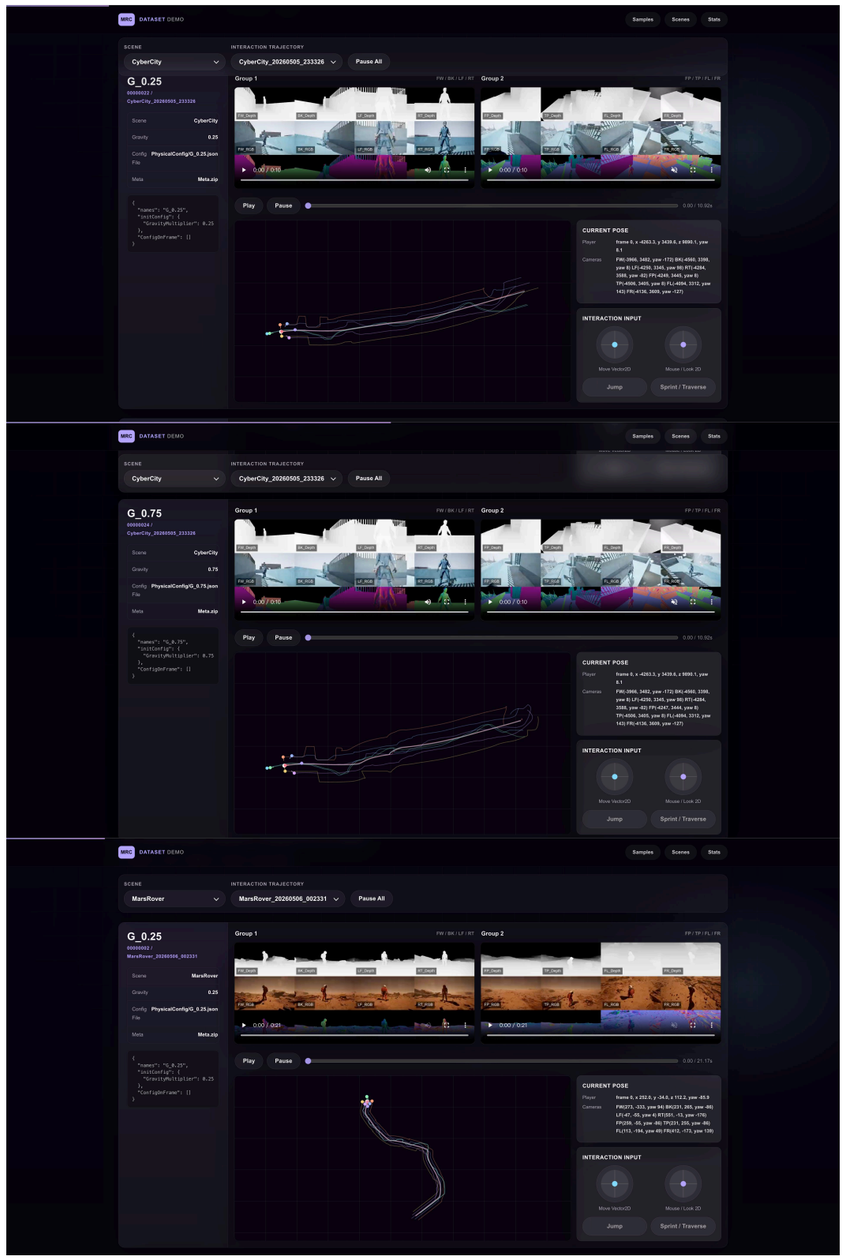

图 11：更多 PhysicsEdit 数据

Figure 12: More PhysicsEdit-Data with other physics config like friction, bounce coefficient

图 12：更多 PhysicsEdit 数据，包含摩擦、弹性系数等其他物理配置

Figure 13: More PhysicsEdit-Data with other physics config like friction. bounce coefficient

图 13：更多 PhysicsEdit 数据，包含摩擦、弹性系数等其他物理配置